WASHOE COUNTY URBAN STORMWATER MANAGEMENT PROGRAM

VOLUME II
STREET PARTICULATE DATA COLLECTION AND ANALYSES

Prepared for
WASHOE COUNCIL OF GOVERNMENTS

Prepared by
CH2M HILL
ROBERT PITT
COOPER AND ASSOCIATES, INC.
CONSULTING ENGINEERING SERVICES, INC.

August 1982

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URBAN STORMWATER
MANAGEMENT PROGRAM

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ROBERT PITT
CH2M HILL

Robert Pitt, Principal Investigator
Roger Sutherland, Project Manager (CH2M HILL)

CH2M HILL
engineers
planners
economists
scientists

R14934.A0.02

Mr. Leonard E. Crowe
Washoe Council of Governments
P.O. Box 1331
Reno, Nevada 89504

Dear Mr. Crowe:

Subject: WCOG Urban Stormwater Management Program
Task 2 Report

In accordance with our contract dated May 4, 1981, subsequent correspondence, and discussions, and the comments concerning our August 1982 draft report, we are submitting, in association with Robert Pitt, the subject report. Since this report is the second of four documents to be provided, it has been assigned a Volume II designation.

Sincerely,

Roger C. Sutherland, P.E.
Project Manager

433.031.1

Portland Office
2020 S.W. Fourth Avenue, 2nd Floor, Portland, Oregon 97201 503/224-9190 TELEX: 360103 CH2M PTL

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Chapter 1

Chapter 1
INTRODUCTION

The purpose of this report, which is the second of four volumes concerning a comprehensive stormwater, air quality, and aesthetics management program, is to describe and present the results of special field studies conducted as part of the program. These special field studies were designed to achieve two major objectives.

The first objective was to obtain data on the site-specific physical and chemical characteristics of the street particulate material accumulating in the Truckee Meadows area. Research on urban stormwater quality conducted in the past decade has identified street particulate material to be the primary source of pollutants found in urban runoff. Data obtained in these field studies will be used to calibrate a stormwater quality model which will, in turn, be used to project both existing and future urban stormwater discharges throughout the Truckee Meadows study area. Both the model calibration and the stormwater discharge projections will be presented in the third volume report.

The second objective of these field studies was to obtain data on the site-specific street-cleaner performance characteristics throughout the Truckee Meadows area. Street cleaning is one of the public works practices that will be described and evaluated in the fourth and final volume report. Street cleaning is an important practice since it can be used to address all three objectives of the management program--stormwater quality, air quality, and aesthetics.

Several features of the Reno and Sparks street cleaning programs that are important to the design of the field studies include:

- The current street cleaning programs are extensive (street cleaning is conducted on a daily to weekly basis in Reno and on a daily to monthly basis in Sparks).
- The public works departments are considering future street cleaning fleet expansions.
- Past repaving of the streets has caused a lip at the curb/street interface at some streets that reduces street cleaning effectiveness.
- The large number of parked cars in many of the commercial and institutional areas also reduces street cleaning effectiveness.
- The cities are currently experiencing equipment maintenance problems.
- In order to meet the frequent street cleaning schedule with existing street cleaning equipment, the equipment is being operated at excessive speeds.

The field studies were designed to address these and other special issues. The sampling program studied street surface particulate accumulation rate characteristics at ten sites throughout the Truckee Meadows in both Reno and Sparks, Nevada. The chemical characteristics of the particulate material at each site were determined. Most of the sites included street cleaner productivity (i.e.,

performance) tests. The field sampling program began in June 1981 and ended in September 1981.

A series of accumulation samples were collected at the test sites during the 4-month program. Samples were also taken from only the driving lane portion of the streets during the second test period. The initial multiple pass street cleaning effort (before the accumulation tests) were monitored at some sites to obtain an indication of extensive multiple pass street cleaning effectiveness. Other street cleaning productivity tests were conducted in some areas to examine the relationships between initial street surface loadings and residual street surface loadings for single street cleaning passes. Other tests were also conducted to examine various vehicle operating speeds, across-the-street redistribution of particulates during street cleaning, and different equipment types. The driving lane and across-the-street tests result in data pertaining to parked car problems and lipped curb/street interface problems.

This report describes the data collection program and summarizes the data obtained. The Truckee Meadows data are also compared to data obtained elsewhere using similar field procedures.

Chapter 2

Chapter 2
DATA COLLECTION

2.1 SITE SELECTION

Twelve candidate sites were visited in early May 1982. Figure 2.1 shows the approximate location of each site. The figure also shows the various watershed boundaries and their number designations. These watershed numbers correspond to those used by Comarc Design Systems (1978) in the development of land use data summaries. The locations of the three urban stormwater quality and quantity sampling stations are also shown on the figure along with their respective watershed boundaries.

Table 2.1 summarizes the important characteristics of the twelve candidate sites for street particulate sampling. All sites examined had gutters on each side of the street and were served by closed pipe sewerage systems. Sites 3, 4, 5, and 8 are located within the watershed boundaries of one of the three stormwater quality and quantity sampling stations. Site 3 is a commercial strip site located within the Vine Street Drain area. Sites 4, 5, and 8 are medium density residential sites of somewhat similar characteristics. Site 4 is located within the Vine Street Drain area and Sites 5 and 8 are located within the Casazza Drain area and the McCarran Drain area, respectively.

The development ages for these three stormwater sampling areas reflect the complete age range of urban development in the region. Casazza is the oldest area, built between 1950 and 1965. Vine was developed between 1966 to 1979; the McCarran area is the newest, built between 1980 and 1981.

Sites 7 and 9 are also medium density residential areas. However, site 7 is in a low income area, while site 9 is in a new development area. In addition, site 9 is the only study area that has good street surface conditions with a relatively smooth surface texture. Site 1 was studied because it is an important commercial central business district area with a high parking density. Site 2 was not studied because it is similar to site 1. Site 6 represents a hospital/institution area with many parked cars. Site 10 was not studied because it is a high density residential area with a medium income level and was represented by the other residential study areas. Site 11 is an industrial site with high truck volumes. Site 12 was studied because it is a high density residential area with a low income level.

These ten sites (i.e., 1, 3, 4, 5, 6, 7, 8, 9, 11, and 12) represent most of the conditions that may be encountered in the whole Truckee Meadows area. The analyses of the accumulation rate data collected from these areas were used to estimate complete basin conditions.

In addition to the accumulation tests conducted in each of these ten locations, many of the sites were also used to study street cleaner productivity. The specific tests conducted in each study area are described in Section 2.2.

These site conditions were carefully examined to identify the important characteristics that affect the street dirt accumulation rates and cleaning productivities. The most important factor appears to be the interacting effect of street texture and street/gutter interface conditions. As described in Chapter 3, smooth streets with smooth street/gutter interfaces are the cleanest and

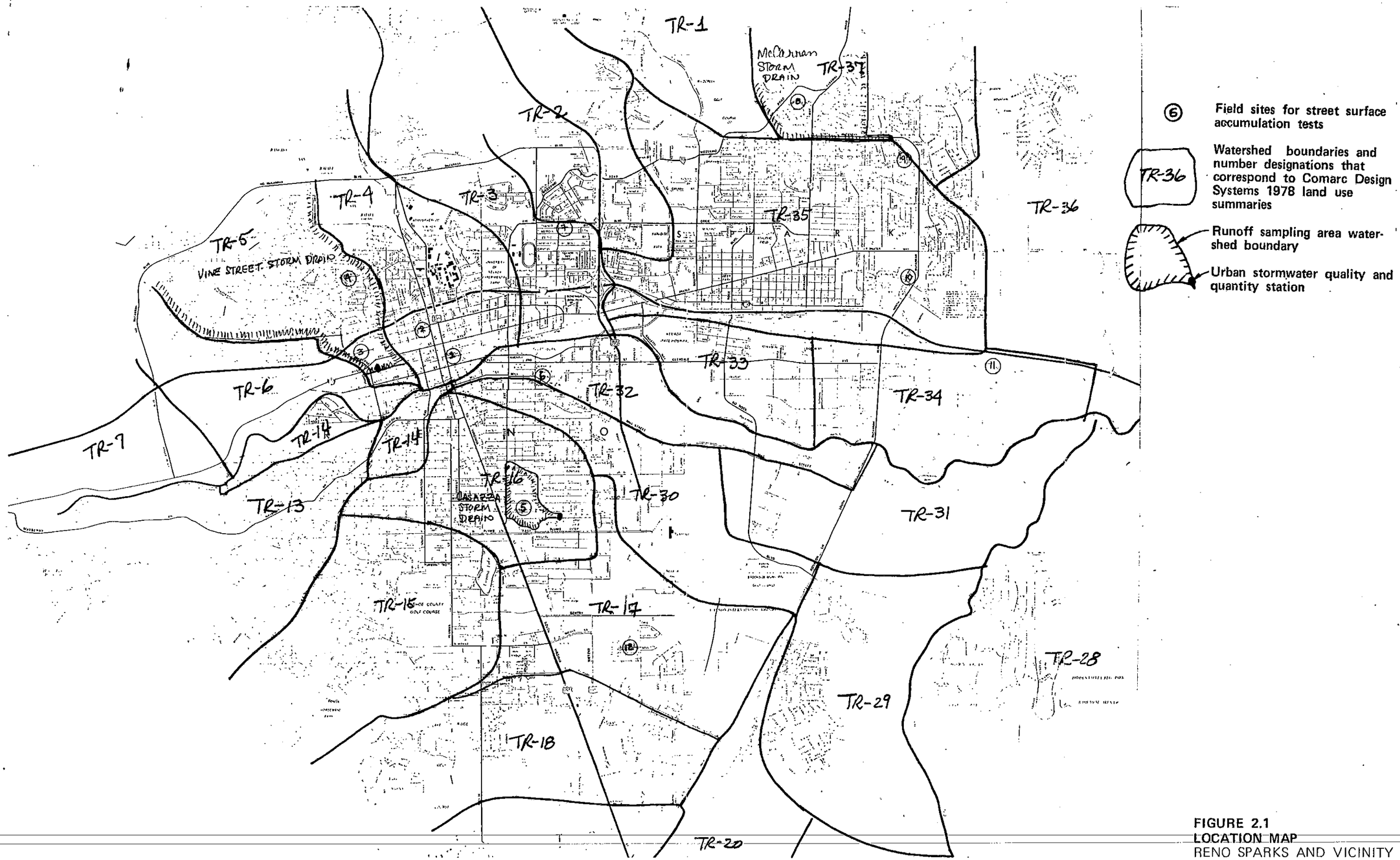

FIGURE 2.1
LOCATION MAP
RENO SPARKS AND VICINITY

Table 2.1

SITE SELECTION SUMMARY INFORMATION (RENO/SPARKS)

Characteristic	Site 1	2	3	4	5	6
Location	E. Commercial near Virginia (CBD)	Arlington at 6th	5th at Keystone	Williams and Brookfield	Locust between Colorado and Grand Canyon	Mill St. near Kirman
Drainage basin	TR-4	TR-4	TR-5 (Vine St.)	TR-5 (Vine St.)	TR-16 (Casazza)	TR-32
Income level (if residential)	-	-	-	Median	Median	-
Age of development (construction date)	1966-79	1980-81	1950-65	1966-79	1950-65	1966-79
Gutter material[a]	C	C	A	C	C	C
Curb shape[b]	S	S	S	R	S	S
Gutter condition[c]	G	G	PF	PF	F	G
Gutter/street interface[d]	S	S	S	L	L	S
Street condition[c]	FG	PF	FG	F	FG	G
Street texture[e]	S	I	S	R	I	R
Number of parking lanes	2	2	0	2	2	2
Number of driving lanes	4	2, 4	5	2	2	4
Vegetation density[f]	N	H	N	M	M	H
Vegetation type[g]	-	D	-	D	D	D
Parking density[h]	H	H	N	L	M	H
Catchbasin density (per intersection)	4	2	2	Low	2	2
Traffic volume	H	H	H	L	M	H
Truck volume	L	L	M	L	L	L
Traffic speed (mph)	30-40	30-40	Over 40	30-40	30-40	30-40
Street slopes (%)	Under 1%	Under 1%	1-3%	Over 3%	Under 1%	Under 1%
Nearby sediment source?	No	No	No	No	Yes	No
Selected for study?	Yes	No	Yes	Yes	Yes	Yes
Across-the-street second phase?	Yes	No	Yes	No	No	No

[a] A = asphalt; C = concrete.
[b] S = Straight; R = rolled.
[c] P = poor; F = fair; G = good.
[d] S = straight; L = lipped.
[e] R = rough; I = intermediate; S = smooth.
[f] N = none; L = low; M = medium; H = high.
[g] D = deciduous; M = mixed; E = evergreen.
[h] N = none; M = medium; H = high.

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Table 2.1 (continued)

Characteristic	Site 7	8	9	10	11	12
Location	Trainer near Montello	Lepori Way near Wilma Ct.	Tallman near Allard Ln.	Pine Meadows at Lincoln Way	Meredith Way near Glendale Ave.	Parkview St. at Mazzone
Drainage basin	TR-3	TR-37 (McCarran)	TR-35	TR-35	TR-34	TR-17
Income level (if residential)	Low	Median	Median	Median	-	Low
Age of development (construction date)	1950-65	1980-81	1981	1966-79	1966-79	1966-79
Gutter material[a]	C	C	C	C	C	C
Curb shape[b]	S	S	S	S	R	S
Gutter condition[c]	G	G	G	G	G	PF
Gutter/street interface[d]	S	S	S	S	S	L
Street condition[c]	FG	G	G	G	FG	F
Street texture[e]	I	R	S	R	R	S
Number of parking lanes	2	2	2	2	0	2
Number of driving lanes	2	2	2	2	4	2
Vegetation density[f]	M	L	N	L	L	N
Vegetation type[g]	D	M	-	D	D	-
Parking density[h]	M	L	L	M	N	M
Catchbasin density (per intersection)	2	2	Very low	1	Low	Low
Traffic volume	L	L	L	L	H	M
Truck volume	L	L	L	L	M	L
Traffic speed (mph)	30-40	30-40	30-40	Under 30	30-40	30-40
Street slopes (%)	Under 1%	1-3%	Under 1%	Under 1%	Under 1%	Under 1%
Nearby sediment source?	No	Yes	Yes	Yes	Yes	Yes
Selected for study?	Yes	Yes	Yes	No	Yes	Yes
Across-the-street second phase?	No	Yes	No	No	Yes	Yes

[a] A = asphalt; C = concrete.
[b] S = straight; R = rolled.
[c] P = poor; F = fair; G = good.
[d] S = straight; L = lipped.
[e] R = rough; I = intermediate; S = smooth.
[f] N = none; L = low; M = medium; H = high.
[g] D = deciduous; M = mixed; E = evergreen.
[h] N = None; M = medium; H = high.

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"easiest" to clean. Driving lane street dirt loadings are also much less than the loadings found in the first eight feet from the gutter, which is the area commonly cleaned by street cleaners. Streets with rough textures and lipped or cracked street/gutter interfaces have the opposite characteristics. High traffic volumes on rough streets (such as Mill Street, site No. 6) tend to reduce the expected driving lane street dirt loadings. Table 2.2 groups the test locations according to these important characteristics.

As part of this study, the complete Truckee Meadows basin was examined by drainage area to obtain the distribution of these characteristics. Table 2.3 summarizes these drainage basin characteristics. These 23 drainage basins comprise almost all of the existing developed areas in Reno and Sparks. About 225 locations were surveyed by Washoe COG staff for street conditions and other important factors. Table 2.3 shows that most of the streets have smooth or intermediate textures, but almost one-third of the streets have lipped street/gutter interfaces. Many of the streets also have nearby (within 15 feet) sediment sources. The street cleaning program recommendations contained in the Volume IV report for this project address each of these drainage basins separately, based on the observed characteristics found in the test basins during the field studies.

Table 2.2

TEST AREA GROUPINGS

Street Surface Texture	Street/Gutter Interface	Sites
Smooth	Smooth	CBD,[a] Fifth,[a], Tallman
Smooth	Lipped	Parkview
Intermediate	Smooth	Trainer
Intermediate	Lipped	Locust
Rough	Smooth	Lepori, Glendale, Mill[a]
Rough	Lipped	Brookfield

[a]These areas also have high traffic volumes.

Table 2.3

TRUCKEE MEADOWS DRAINAGE BASIN
STREET CHARACTERISTICS

TR Basin	Gutter/Street Interface: Smooth	Gutter/Street Interface: Lipped	Street Texture: Smooth	Street Texture: Intermediate	Street Texture: Rough	Street Texture/Interface Condition: Smooth/Smooth	Smooth/Lipped	Intermediate Smooth	Intermediate Lipped	Rough/Smooth	Rough/Lipped	Street Condition: Good	Fair	Poor	Nearby Sediment Source?: Yes	No	High Traffic Volume	Number of Responses for the TR Basin	Total Area of Basin (acres)
2	71%	29%	13%	63%	25%	14%	-	43%	29%	14%	-	56%	33%	11%	44%	56%	-	9	1,621
3	71	29	13	74	13	15	-	57	29	-	-	50	50	-	37	63	13%	8	2,412
4	44	56	35	59	6	19	19%	25	31	-	6%	35	53	12	12	88	35	17	3,840
5	69	31	4	88	8	4	-	58	31	8	-	38	62	-	-	100	15	26	5,353
6	50	50	17	83	-	-	17	50	33	-	-	17	83	-	-	100	33	6	1,425
13	73	27	36	64	-	36	-	36	27	-	-	64	36	-	36	64	-	11	1,257
14	-	100	25	75	-	-	25	-	75	-	-	50	50	-	25	75	-	4	442
15	58	42	31	69	-	25	8	33	33	-	-	62	38	-	23	77	-	13	3,043
16	13	88	25	75	-	-	25	13	63	-	-	56	44	-	12	88	-	16	1,202
17	75	25	15	62	23	17	-	42	25	17	-	46	54	-	31	69	23	13	1,771
18	91	9	8	85	8	9	-	73	9	9	-	69	31	-	23	77	-	13	2,368
19	50	50	-	50	50	-	-	-	50	50	-	-	100	-	50	50	-	2	6,584
20	100	-	-	33	67	-	-	100	-	-	-	33	33	33	100	-	33	3	10,414
28	100	-	-	33	67	-	-	-	-	100	-	-	67	33	100	-	-	3	3,567
29	100	-	-	100	-	-	-	100	-	-	-	100	-	-	20	80	-	5	1,811
30	86	14	-	100	-	-	-	86	14	-	-	57	43	-	71	29	29	7	1,366
31	100	-	-	100	-	-	-	100	-	-	-	-	100	-	100	-	-	2	1,605
32	60	40	60	40	-	40	20	20	20	-	-	80	20	-	83	17	20	5	855
33	67	33	-	86	14	-	-	50	33	17	-	71	14	14	71	29	13	7	1,124
34	100	-	-	80	20	-	-	80	-	20	-	80	20	-	100	-	-	5	1,446
35	79	21	13	82	5	14	-	58	22	6	-	69	31	-	10	90	-	39	2,827
36	100	-	-	100	-	-	-	100	-	-	-	100	-	-	-	100	-	3	7,440
37	71	29	11	78	11	11	-	67	11	11	-	100	-	-	-	100	-	9	1,523
Total[a]	68	32	16	75	9	11	5	49	27	7	<1	57	40	3	26	74	12	226	65,296

[a]Weighted by responses.

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2.2 PROGRAM DESIGN

Accumulation tests involved collecting many half street (curb to center of the street) street surface subsamples throughout each individual site area. About 15 of these subsampling strips were collected over about a single curb-mile of the area. Each site was cleaned with three passes at the beginning of each of the two test cycles, and accumulation samples were then collected at various times following this initial cleaning: about one day after the area had been cleaned, another sample about three days after the site was cleaned, and then another sample about seven days after it was cleaned. Several more samples were then collected on a weekly basis in each area. After about six weeks, five of the sites were again cleaned with three passes and the cycle repeated a second time. This resulted in about 15 street surface particulate samples collected in many of the areas over the three month period of time. These data points were then plotted and accumulation curves prepared, based upon these individual street surface particulate loading values.

During the second sampling period in five test areas, special center street samples were collected immediately after the full street width samples were collected. These samples only represented the driving lane strip, starting about eight feet from the curb on one side of the street to the street's center. About eight of these samples were collected in each of the five second period study locations. They indicate the distribution of the street surface materials across the street before and after cleaning. This distribution varies significantly depending upon the specific parking conditions and street textures at the sites. This information will be used in assessing the use of full street width street cleaning operations and the effectiveness of parked car controls.

Several types of street cleaner productivity tests were also conducted. A test of the effectiveness of multiple passes (three) was conducted during the cleanup phases at most of the selected accumulation test sites. A street surface sample was collected from the sites before they were cleaned and then again as soon as possible after the sites had been cleaned. This resulted in about 14 sets of data from the test sites. During the second phase, samples were collected between each of three adjacent street cleaning passes at five sites. This resulted in single pass productivity values.

An evaluation of cleaning the full width of the street was made, based upon examining the results from the accumulation tests and from the full-width street cleaning tests. Several days of testing in three sites were used to examine different vehicle speeds , different equipment types (a mechanical Mobil versus a vacuum-assisted Ecolotec), the effects of full street width cleaning, and the redistribution of particulates across the street during street cleaning.

Accumulation Studies

Ten sites were selected in Reno and Sparks that represented the range of conditions to be found throughout the Truckee Meadows basin. These sites were described in Section 2.1. All ten sites were cleaned three times in a short

period of time in early June, 1981. Street surface particulate samples were obtained during the next six weeks in order to determine the accumulation rate of street dirt as a function of time since last cleaned or major rain. Five of the sites (the central business district, Fifth St., Lepori, Glendale, and Parkview) were selected for continued testing. These sites were extensively cleaned in mid August, 1981, and street surface samples were collected until the first part of October, 1981.

The second series of tests also included street dirt samples that represented loadings from only the driving lane portions of the streets. These data were used to examine the potential benefits of full street width cleaning, and the effects of parked cars and street texture on street dirt distribution. These extra samples were obtained immediately after the full street width samples were obtained and at similar subsampling strip locations.

Table 2.4 summarizes the numbers of samples obtained at each sampling location. These 215 samples were processed in the soils laboratory and chemical laboratory, as described in Section 2.4. Appendix A contains the observed street dirt loadings, by particle size and constituent for each sample. Appendix B shows the observed chemical concentrations for each particle size and for each sample. These data were used to calculate the accumulation rates for each site and time period. Typical street surface loadings were also determined for each site. Sites with similar loadings, accumulation rates, and chemical strength characteristics were identified and grouped. Factors which influence these values were identified for use in analyzing complete Truckee Meadows basin street dirt characteristics, by drainage basin.

Street Cleaning Productivity Studies

Several types of street cleaning studies were conducted in Reno and Sparks. The initial extensive cleaning effort at the beginning of the first study period included street dirt samples at the beginning and end of the three cleaning passes throughout the whole study site areas. The extensive cleaning effort at the beginning of the second study period included street dirt samples at the beginning and end of the three cleaning passes and samples between each pass. These samples therefore resulted in data describing the cleaning effectiveness of single and multiple passes for most of the study sites. The street dirt samples collected for these tests were obtained in the same manner as the accumulation tests, and represented complete across-the-street conditions. Other samples at the beginning of the second test period were also obtained to only represent driving lane street cleaning productivity conditions. Table 2.5 summarizes the sampling effort for these full scale street cleaning tests. These 39 pairs of samples were used to compare initial (before cleaning) to residual (after cleaning) street dirt loading conditions for different categories of street texture and street/gutter interface conditions. Most of the full scale street cleaning test data was obtained at the sites selected for the second test period sampling.

Other street cleaning tests were conducted at the end of July, 1981 in parts of three of the sites: Brookfield, Glendale, and Lepori. Other tests were also conducted in parts of the Brookfield area in October, 1981. These small-scale street cleaning productivity tests examined special features of Reno and Sparks conditions and were conducted over a several block length of the three study areas. These tests examining cleaning productivities for different vehicle

Table 2.4

ACCUMULATION SAMPLING EFFORT

Site	Number of Samples Obtained			
	First Period Full-Width	Second Period Full-Width	Second Period Driving Lane Only	Total
Parkview	8	13	13	34
Mill	8	0	0	8
Fifth	11	13	13	37
Central Business District	10	10	9	29
Tallman	9	0	0	9
Trainer	9	0	0	9
Lepori	10	13	13	36
Locust	10	0	0	10
Brookfield	9	0	0	9
Glendale	8	13	13	34
Total	92	62	61	215

Table 2.5

FULL SCALE STREET CLEANING TESTS

Site	Number of Tests Conducted			
	Full-Width Single Pass	Driving Lane Single Pass	Multiple (3) Pass Full-Width	Total
Parkview	3	3	2	8
Mill	4	3	2	9
Fifth	0	0	1	1
Central Business District	1	0	1	2
Tallman	0	0	1	1
Trainer	0	0	1	1
Lepori	3	3	2	8
Locust	0	0	1	1
Brookfield	0	0	1	1
Glendale	3	3	1	7
Total	14	12	13	39

speeds, the differences in productivities for different street cleaning equipment types, and the redistribution of street dirt across the entire street after curb-side cleaning and full width street cleaning.

The speed tests involved identifying four adjacent curb lengths, each several hundred yards long. Initial street dirt samples (made up of ten subsamples each) were obtained from each section. Each section was then cleaned at a different vehicle speed (about 15, 8, 4, and 2 miles per hour) and residual samples were then collected.

The effects of different vehicle types were also studied using a similar procedure. Three curb sections, each several hundred yards long, were identified in the Glendale area. Initial samples (made up of ten subsamples) were obtained in each section. An Ecolotec, a vacuum-assisted mechanical street cleaner, cleaned the first strip; a Mobil, the "standard" mechanical street cleaner that was used in the other tests, then cleaned all three curb lengths. The first section was therefore cleaned twice, first by the Ecolotec and then by the Mobil. The two other sections were each cleaned once by the Mobil. Street dirt samples were collected after the first Ecolotec pass in the first section and after the Mobil in each of the three passes. The samples represented full street width loading conditions.

The across-the-street tests were conducted in smaller areas in the three test areas. Three adjacent curb sections, each only five or ten feet long, were sub-divided and marked into strips at different distances from the curb. Table 2.6 shows the strip widths that were used. The curb interface was always located in the second strip. The third section was completely vacuumed, by strip, before street cleaning. The street cleaner made a single pass next to the curb, after which the second section was completely vacuumed, by strip. Finally, the center of the street was cleaned, and the first section was completely vacuumed, also by strip. This information was used to see how street dirt was re-distributed across the street during street cleaning operations, both for normal curb cleaning and for unusual full width street cleaning.

Table 2.7 summarizes the testing effort for these small scale street cleaning tests. These 24 tests included many samples, as each across-the-street test involved ten or 12 samples and the other tests included two samples. Therefore, about 86 samples were processed for these small scale tests and about 78 samples were processed for the full-scale productivity tests. The samples were processed as described in Section 2.3. Appendix A includes loading values for each full-scale test along with the accumulation loading values. The summary sheets for the small scale tests are included in Appendix C. No chemical analyses were performed on the small-scale test street dirt samples.

Table 2.6

ACROSS-THE-STREET STRIP WIDTHS, AS MEASURED
FROM THE CURB FACE TOWARD THE CENTERLINE
(feet)

Brookfield	Glendale	Lepori
0-1	0-1½	0-1
1-2	1½-3	1-2
2-4	3-8	2-5
4-8	8-16	8-14
8-12	16-25	8-14
12-17		14-20½

Table 2.7

SMALL-SCALE STREET CLEANING TEST EFFORT

Site	Number of Tests Conducted			
	Speed Effects	Equipment Types	Across-the-Street Redistribution	Total
Brookfield	8	0	2	10
Glendale	4	4	1	9
Lepori	4	0	1	5
Total	16	4	4	24

2.3 STREET SURFACE PARTICULATE SAMPLING PROCEDURES

Special Features of Reno/Sparks Sampling Program

Street surface sampling procedures developed previously for a San Jose street cleaning project (Pitt,1979) were used in collecting the street surface samples for this project. This project used a Pullman/Holt three-horsepower vacuum. This large vacuum with its two-inch diameter hose and special modified gulper attachment simplified street surface sampling. A single vacuum with this size motorhead and hose diameter is adequate in collecting representative street surface samples. During the San Jose study, 1-1/2 inch hoses and attachments were used with two separate two-horsepower vacuum canisters connected with a "Y" connector. The double vacuums required extra time to empty and transfer the samples.

All of the strip samples in Reno and Sparks were collected with a pulling action starting from the middle of the street and going towards the curb. This increased the safety of the operator, as he could then monitor traffic conditions; and it also minimized traffic obstructions. The streets in Reno and Sparks are too rough for the faster pushing action. The streets are also sufficiently dirty to allow adequate samples to be collected from half of the street (from the central line to the curb). Safety precautions were stressed; the sampler constantly observed oncoming traffic and monitored the collection efficiency of the sampler with periodic glances at the street surface and by listening to the noise of the rocks coming up the vacuum wand and through the hose. The use of the generator and the vacuum in the back of the flatbed truck also simplified the sample collection operations and made the unit more maneuverable.

Sample Collection

Each subsample is supposed to include all of the street surface material that would be removed during a severe rain (including loose materials and caked-on mud in the gutter and street areas). The locations of the subsample strips were selected to ensure that they had no unusual loading conditions. For example, a subsample was not collected through the middle of a pile of leaves. It was collected where the leaves were lying on the street in their normal distribution pattern. When possible, wet areas were also avoided. Each subsample was collected in a narrow strip about six inches wide (the width of the gulper) from the center of the street to the curb.

When sampling, the leading edge of the gulper was slightly elevated above the street surface (1/8 inch) to permit pebbles and large particles to enter. The gulper was lifted to accept larger material as necessary. A large accumulation of leaves in the subsample strip was manually picked up and placed in the sample storage container in order to prevent the hose from clogging.

An important aspect of the sample collection was the speed at which the gulper was moved across the street. A very rapid movement significantly decreased the amount of material collected. Too slow a movement required more time than was necessary. The correct movement rate depended on the roughness of the street and the amount of material on it. When sampling a street that had a heavy loading of particulates or a rough surface, the wand should travel at a velocity of less than one foot per second. In areas of lower loadings and smoother streets, the wand could travel at a velocity of two or three feet per second. The best indication of the correct collection speed was made by a visual determination of how well the street was being cleaned in the sampling strip and by listening to the collected samples rattling up the wand and through the vacuum hose. The objective was to remove everything that was lying on the street that could be removed by a significant rainstorm. It was quite common to leave a visually cleaner strip on the street where the subsample was collected, even on streets that appeared to be clean.

When moving from one subsample location to another, the hose, wand, and gulper were securely placed in the truck. The hose was placed so as not to touch the generator's muffler, which could burn a hole in it. The generator and the vacuum were left on and in the truck during the entire subsample collection period.

Several hundred grams of sample material were needed for the laboratory tests. An after-cleaning subsample was not collected from exactly the same location as the before-cleaning subsample. They were taken from the same general area, but at least a few feet apart.

A record of information which may later be necessary to resolve disputes and inconsistencies regarding times, dates, weights, and sample numbers was also kept. As each subsample was collected, the corresponding number for the subsample was checked off on a tally sheet. The time at the beginning and the end of the sampling run was also noted. The weather conditions during sampling was also noted, along with any unusual circumstances.

Sample Transfer

After all the subsamples for a test area were collected, the hose and wand were cleaned. Leaves and rocks that may have become caught were carefully removed and placed in the vacuum can. The generator was then turned off. The vacuum was either emptied at the last station or at a more convenient location.

To empty the vacuum, the top motor unit was removed and placed out of the way of traffic. The secondary, coarse vacuum filter was removed form the vacuum can, and using a small whisk broom, was carefully brushed into a large funnel placed in the storage container. The primary dacron filter bag was kept in the vacuum can and shaken carefully to knock off most of the filtered material. (The hose inlet was blocked with a leg or knee and the primary filter bag was held onto the drum with arms and chest.)

The dust inside the can was allowed to settle for a few minutes, then the primary filter was removed and brushed carefully into the sample container with the whisk broom. Any dirt from the top part of the bag where the bag was bent over the top of the vacuum was also carefully removed and placed into the sample

container.

After the filter was removed and cleaned, the vacuum can was tilted onto the large funnel on top of the sample container. The inside of the vacuum can was brushed out with a soft three or four inch paint brush to remove the collected sample. In order to prevent excessive dust losses, the emptying and brushing was done in areas protected from the wind. To prevent inhaling the toxic sample dust, mouth and nose dust filters were worn while removing the samples from the vacuum.

The sample storage containers were labeled with the date, the test area's name, and an indication of whether the sample was taken before or after street cleaning or if it was an accumulation (or other type of) sample. Finally, the lids of the sample containers were taped shut and transported to the soil laboratory for analyses.

2.4 LABORATORY PROCEDURES

Mechanical analyses were performed on the street dirt samples (total sample weight and a particle size breakdown into eight particle sizes). Composite samples for each of the test basins and study periods were sent to a commercial laboratory (Sierra Environmental Laboratory, Reno, Nevada) for chemical analyses. In addition, each sample was also chemically analyzed, before separating into particle sizes. The particle size groups were <63, 63 to 125, 125 to 250, 250 to 500, 500 to 1000, 1000 to 2000, 2000 to 6370, and >6370 microns. All samples were chemically analyzed for total solids, median particle size, cadmium, chromium, lead, zinc, chemical oxygen demand, total phosphorus, nitrate nitrogen, nitrite nitrogen, and total Kjeldahl nitrogen (organic nitrogen plus ammonia). This section describes the sample check-in, mechanical analysis, and chemical analysis procedures.

Sample Check-in

When the samples are returned to the laboratory, each sample was given an identification number which was written on the sample container and on the sample checklist. Each container was also weighed as it was received. The average storage container tare weight was then subtracted and the resultant sample net wet weight was noted on the checklist. This information may be needed to resolve future problems that may occur when handling so many samples.

Mechanical Analyses

Most of the street surface samples were dry when they were collected. Most of the typically monitored street dirt constituents and characteristics tend to remain fairly stable (i.e., they do not undergo significant physical, chemical, or biological changes during the short periods of storage). In fact, street surface particulates may lie exposed to a wide range of climatic conditions on the road surface for several months before rains wash them into the receiving waters. The samples were stored before analyses, although the storage times were kept to a minimum. Some storage was necessary because of the need to composite samples over the testing periods.

The samples were transferred from their storage containers to labeled drying pans. These pans were then placed in a low-temperature drying oven for several hours. As the drying temperature was held to less than 120 degrees F., this heating did not affect the samples' chemical characteristics.

The gross and net weights of the samples were measured and the percent moisture calculated.

After adequate drying, the samples were passed through a coarse screen made of 1/4 inch (6370 microns) hardware cloth attached to a wooden frame about two feet square. The coarse material retained on the screen was weighed, put in a plastic bag, and placed in storage for later chemical analysis (in a manner

similar to all of the other size fractions). The remaining portion of the sample (smaller than 6370 microns) was placed in a set of mechanical, stainless steel sieves for size separating. The sieve sizes used were 63, 125, 250, 500, 1000, and 2000 microns. Fractions smaller than 63 microns were retained in the pan on the bottom of the sieve stack.

The gross dry weight of each sieve, the associated retained sample weight, and the tare weight of the sieve was entered on the sieve forms. The net dry weight of this sample was then calculated and the result totaled. The percentage of sample in each size fraction was also calculated and reported on this sieve form, along with the calculated loading in pounds per curb mile.

Placing the sample information on the log sheets, drying sheets, and sieve sheets is important for quality control. Detailed sample descriptions entered on these sheets included; sample I.D. number, date sampled and analyzed, study area, and the net raw sample weight. This weight was compared with the total net dry weight for the sample (as a cross check). The net raw weight would not be expected to be as precise as the total of the net sieved dry weights (because of the weight lost due to drying and the fact that more precise scales were used to weigh the sieves), but the difference should be less than about five percent. If a large discrepancy existed between these two weight values, then the sample was rechecked by reviewing the log sheets and forms, and reprocessed if necessary.

Sample Compositing for Chemical Analyses

The chemical concentrations of the measured constituents can vary substantially among particle size ranges, as described in Chapter 3. It is therefore important to analyze different particle sizes for the different chemical constituents. Chemical analyses were also performed on each separate whole sample, after recombining according to the sample particle size breakdown. Compositing involved combining equal quantities (typically five or ten grams) of each size fraction of all samples collected in a single study area, for each sampling period. Equal quantities of samples were combined for the composites, because we were interested in determining a time-averaged chemical concentration of the material.

Complete sample chemical analyses were also obtained in order to more accurately estimate the accumulation rates for each street dirt constituent separately. After the samples were dried and sieved, a small amount of each sample was weighed, according to the specific particle size distribution, and recombined. This was much more accurate than splitting the complete sample before sieving because of the wide range of particle sizes present (from about ten to 10,000 microns in diameter).

Chemical Analyses

Each composited sample was typically made up of five to 20 subsamples. These composited samples were placed in small sample containers, labeled, and sent to the commercial laboratory (Sierra Environmental Monitoring of Reno) for chemical analyses. The laboratory milled the coarser samples to sizes less than 125 microns before they were analyzed. The heavy metals (lead, chromium, cadmium, and zinc) were analyzed for total metal content using procedures described in Standard Methods for the Examination of Water and Wastewater, 14th edition, Part

301A. The ground samples were digested in a mixture of nitric and hydrochloric acid over heat and were analyzed for each metal using flame atomic absorption spectrophotometry. Chemical oxygen demand was determined using the procedures described in Part 508 of Standard Methods. Total phosphorus was determined by the digestion and ascorbic acid colorimetric procedure given in Methods for Chemical Analysis of Water and Wastes, 1979, USEPA, Method 365.3, Procedure 8.1. Nitrite and nitrate samples were saturated with water and the filtered extract was analyzed following procedures given in Parts 420 and 419D (Brucine method), respectively, of Standard Methods. Total Kjeldahl nitrogen samples were digested according to Part 421.4C of Standard Methods and an ammonia specific ion electrode was used for analysis.

Chapter 3

Chapter 3
DATA ANALYSES

3.1 CLIMATIC CONDITIONS

Climatic conditions greatly influence urban runoff characteristics and public works practices. Frequent and heavy rains result in large volumes of relatively clean runoff. Street surfaces are also kept fairly clean by frequent rains. In an area with infrequent rains, the long dry periods allow much more material to accumulate in the urban catchment. The resultant urban runoff, therefore, usually has greater pollutant concentrations. Street cleaning can be more important as an urban runoff control measure in drier areas, because dirty streets contribute a larger fraction of the total urban runoff yield. The street dirt samples collected during this study were analyzed for accumulation rates (see Section 3.4) and were found to be strongly influenced by wind conditions during the study.

Table 3.1 summarizes the 1981 climatic conditions recorded at the Reno International Airport. The temperatures at the airport ranged from a low of about 16°F in January to a high of about 110°F in July. The nighttime lows were in the seventies during the hot periods in July. The daytime highs during January were in the high forties.

The Truckee Meadows study area is dry, with relatively long periods between rains. The precipitation amounts throughout the study area vary substantially because of topographical differences. These variations will be discussed in the third volume report. The following discussion is based on data recorded at the Reno International Airport; its annual precipitation is one of the lowest in the study area.

Rainfall amounts average only about 0.15 inch per event. Each event lasts an average of approximately 5 hours. About 50 events occur each year. The average annual precipitation at the airport is 7.32 inches. The average rainfall intensity is about 0.03 inch per hour, and the average peak 1-hour intensity for the year is about 0.35 hour. Heavy individual rains do occur, with a peak rain storm of about 1 to 2 inches occurring once a year. Most of the rain during 1981 occurred in January, November, and December. No rain was reported during June and July.

Average monthly wind speeds ranged from a low of 3.7 mph in January to a high of 12.6 mph in July. The annual average wind speed was 7.4 mph for 1981. Periods of high winds were common, with peak gusts of more than 50 mph reported in February, November, and December.

Table 3.2 summarizes peak wind conditions reported at the Reno airport for the 31 years from 1950 through 1980. More than 1 percent of the reported days in January, February, March, and December had average wind speeds greater than 20 mph. May through September rarely had these high winds. The high winds experienced during the July 1981 field studies were therefore unusual. High winds do occur during the winter months on days with daytime temperatures well above freezing.

Table 3.1

1981 RENO CLIMATIC CONDITIONS[1]

	Average Temperatures (°F)		Precipitation			Wind (mph)	
	Maximum	Minimum	Days with >0.01 Inch	Total Rain (in)	Total Snow (in)	Average	Maximum
January	48.3	23.8	7	0.85	3.9	3.7	48
February	52.5	25.1	5	0.21	Trace	6.0	52
March	55.2	28.1	7	0.58	2.2	7.5	38
April	66.8	34.6	3	0.21	Trace	7.6	35
May	73.0	42.0	4	0.57	0	7.7	34
June	87.6	48.9	0	Trace	0	8.8	35
July	105.9	79.5	0	0	0	12.6	31
August	92.8	46.0	2	0.36	0	6.4	27
September	84.5	44.9	2	0.07	0	6.3	28
October	63.6	30.2	6	0.64	1.1	6.1	38
November	54.8	30.1	8	2.13	2.0	7.8	56
December	50.7	27.2	7	1.05	0.1	8.4	50
Average Monthly	69.6	38.4	4.3	0.56	0.8	7.4	39
Total	-	-	51	6.67	9.3	-	-

[1]Recorded at Reno International Airport

433.011.1

Table 3.2

RENO WIND CONDITIONS[1]

	Average Hours/Month		Percent of
	>20 mph	>30 mph	Days >20 mph
January	17.0	0.8	2.2
February	15.8	<0.1	1.0
March	34.2	1.8	1.3
April	25.5	<0.1	0.6
May	24.5	0.2	0.1
June	17.5	0.2	<0.1
July	8.5	<0.1	<0.1
August	12.0	<0.1	0.1
September	9.2	<0.1	0.1
October	16.0	1.0	0.3
November	19.5	0.8	0.4
December	25.0	1.0	1.4

[1] Averaged for 1950 through 1980 at Reno International Airport.

Table 3.3 summarizes the winds experienced during the two field study phases of June 8 through August 11, and August 12 through October 1, 1981. The first field phase had winds substantially greater than those during the second phase. Eighteen percent of the days during the first period had average winds greater than 10 mph, as compared with only 4 percent of the days during the second period. Eleven percent of the first period days had peak winds greater than 30 mph, but none of the winds during the second period were of this velocity. As noted in Section 3.4, the first period street dirt loading values were usually greater than those during the second period, possibly because fugitive dust sources contributed to the street loads.

Table 3.3

WIND CONDITIONS DURING STUDY PERIODS

	First Phase (June 8 through August 11)	Second Phase (August 12 through (October 1)
Average daily wind speed (mph)	8.1	6.5
Minimum daily wind speed (mph)	4.0	2.2
Maximum daily wind speed (mph)	17.2	10.5
Average daily maximum wind speed (mph)	22	20
Maximum speed during period (mph)	35	28
Minimum daily maximum wind speed (mph)	10	7
Number of days with average wind speeds >10 mph	12 (18%)	2 (4%)
Number of days with maximum wind speeds >30 mph	7 (11%)	0 (0%)

3.2 STREET DIRT PARTICLE SIZE DISTRIBUTIONS

The particle size characteristics of street dirt significantly affect street cleaner performance and the transport of street dirt in urban runoff. Street cleaners are much more effective in removing larger particles than the generally more polluted smaller particles. The smaller particles are much easier to be suspended and transported in urban runoff. Large particles also settle out in the sewerage and receiving waters more readily than small particles.

Each street dirt sample was sieved into eight different particle sizes before chemical analyses, as described in Section 2.4. These sizes were <63, 63-125, 125-250, 250-500, 500-1000, 1000-2000, 2000-6370, and >6370 microns. Samples representing each time period, study area, and particle size were chemically analyzed. The pollutant characteristics of each particle size are reported in Section 3.3 and in Appendix B.

Table A.2 in Appendix A presents the observed median particle sizes (and pollutant loadings) for all street dirt samples collected for each test area. Median particle sizes for the full street width samples collected during each test period and for the driving lane only samples are shown. The variation in median particle sizes for each grouping is quite small, with relative standard deviations of only about 15 percent. Table 3.4 summarizes these median particle values. The areas having smooth and intermediate textured streets (Parkview, Fifth, CBD, Tallman, Trainer, and Locust) had an average median particle size of about 340 microns during the first test phase and about 330 microns during the second test phase for full street width samples. The driving lane samples during the second phase for these areas had as average median particle size of about 470 microns. The rough textured street samples (from Lepori, Glendale, Mill and Brookfield) had full street particle median sizes of about 520 and 630 microns (for the first and second test phases) and about 780 microns for the driving lane samples. These data show appreciable differences in particle characteristics for different textured street surfaces: rougher streets have larger street dirt particle sizes than smoother streets. The driving lane samples always had larger median particle sizes than corresponding full street width samples. Therefore, the street dirt near the curbs were composed of much finer material than the street dirt found in the center of the roadway.

Table 3.5 shows the observed median particle sizes for samples collected at different distances from the curb, before and after street cleaning. These sites all had rough textured streets and the results agree with the median size data presented in Table 3.4. Some of the near curb samples immediately before street cleaning had large particle sizes, but these large particles were effectively removed by street cleaning. The selective removal of large material in the street cleaning zone (the first eight feet from the curb) therefore reduces the median particle size of the street dirt in this area, especially in areas having frequent street cleaning.

The median street dirt particle size in San Jose (Pitt, 1979) was about 180 microns for smooth asphalt streets, about 250 microns for asphalt streets in poor

Table 3.4

MEDIAN PARTICLE SIZES OF STREET DIRT
FOR TEST AREAS

Test Area	First Phase Full Street Width Sample	Second Phase Full Street Width Sample	Second Phase Driving Lane Sample
Parkview	364	308	513
Fifth	305	307	457
CBD	404	385	453
Lepori	694	630	787
Glendale	549	624	778
Mill	416	-	-
Tallman	367	-	-
Trainer	277	-	-
Locust	340	-	-
Brookfield	408	-	-

Table 3.5

MEDIAN PARTICLE SIZES OF STREET DIRT AS
A FUNCTION OF DISTANCE FROM CURB

Distance From Curb (ft)	Brookfield (7/29/81) Before Cleaning	Brookfield (7/29/81) After Curb-side Cleaning	Brookfield (7/29/81) After Full Street Width Cleaning	Brookfield (10/15/81) Before Cleaning	Brookfield (10/15/81) After Curb-side Cleaning
0→1	375	282	257	813	393
1→2	293	307	257	245	197
2→4	448	378	366	235	188
4→8	542	500	435	502	352
8→12	609	612	550	669	641
12→17	507	494	370	637	649
	Glendale (7/31/81)				
0→1½	1,168	330	460		
1½→3	385	500	500		
3→8	625	470	398		
8→16	816	818	667		
16→25	780	942	811		
	Lepori (7/31/81)				
0→1	410	341	NA*		
1→2	589	560	NA		
2→5	364	389	NA		
5→8	453	514	NA		
8→14	529	593	846		
14→21	440	403	586		

*NA = not available

condition, and about 330 microns for oil and screens overlayed streets. The median particle sizes for Castro Valley street dirt (Pitt and Shawley, 1982) was about 500 microns for asphalt streets in good to fair condition. Street dirt samples recently collected in Ottawa, Ontario (Pitt, 1982) had median particle sizes of about 310 microns for asphalt streets in good to fair condition. The observed median particle sizes for Reno and Sparks were similar to those reported elsewhere for rough streets.

Table A.3 in Appendix A shows the observed total solids loadings separated into the eight particle sizes. This table also shows the percentage of the total solids loadings that occured in each size range, based on loading values for both test periods combined. Table 3.6 summarizes these data for each test area, grouped by street texture. Figure 3.1 is a bar chart comparing these particle size distributions for these texture categories. The smooth and intermediate textured streets have a larger fraction of their total solids loadings in the small particle sizes as compared to the rougher textured streets. The rougher streets, however, have greater percentages of the larger particles. This is similar to observations made in other studies. The rougher streets have more larger particles probably because the rougher street texture is effective in shielding these larger materials from street cleaning and rain washoff. Smooth streets do not offer as good shielding. This lack of larger particles biases the particle size distribution towards the smaller particles, with resultant smaller median particle sizes.

Table 3.6

PARTICLE SIZE DISTRIBUTIONS OF STREET DIRT
(AVERAGES OF ALL FIRST AND SECOND PHASE TEST DATA)

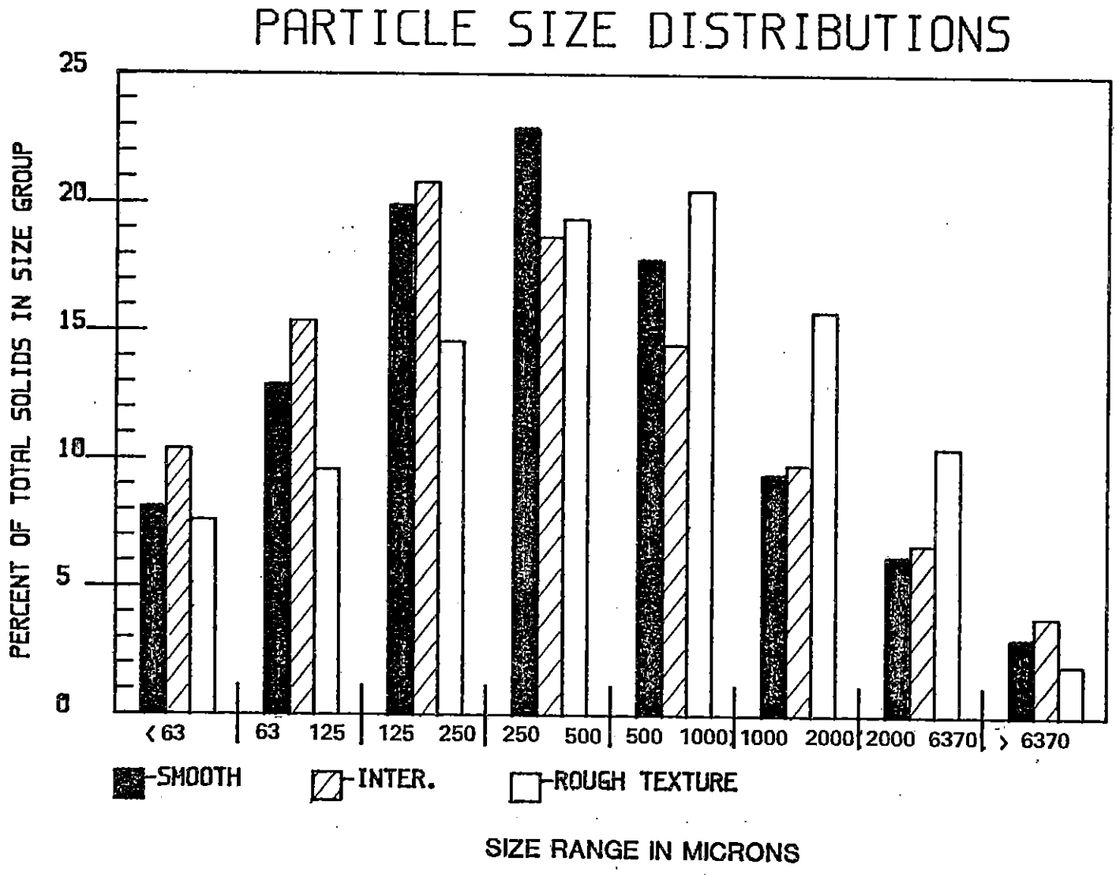
	Particle Size Groups (microns)							
	<63	63→125	125→250	250→500	500→1,000	1,000→2,000	2,000→6,370	>6,370
Mill	6.6%	11.7%	17.0%	22.0%	20.1%	12.0%	9.2%	1.5%
Tallman	10.0	10.9	16.1	28.5	22.2	8.7	3.1	0.7
Trainer	14.1	17.7	19.1	16.5	13.8	9.9	5.7	3.2
Locust	6.6	13.0	22.4	20.9	15.1	9.7	7.7	4.6
Brookfield	9.6	10.7	16.1	21.3	18.4	13.5	8.7	1.7
CBD	6.5	12.5	19.2	21.7	17.1	9.8	7.8	5.4
Parkview	9.1	13.7	20.0	19.0	15.7	10.6	8.3	3.6
Glendale	6.8	8.8	13.5	17.5	21.1	17.1	11.6	3.7
Lepori	7.5	7.3	11.9	16.7	22.5	20.6	12.3	1.1
Fifth	6.6	14.4	24.4	22.4	16.0	8.4	5.7	2.2
Smooth site averages[1]	8.1	12.9	19.9	22.9	17.8	9.4	6.2	3.0
Intermediate texture site averages[2]	10.4	15.4	20.8	18.7	14.5	9.8	6.7	3.9
Rough texture site averages[3]	7.6	9.6	14.6	19.4	20.5	15.8	10.5	2.0

[1] Smooth sites: CBD, Fifth, Tallman, and Parkview.

[2] Intermediate texture sites: Trainer and Locust.

[3] Rough texture sites: Lepori, Glendale, Mill, and Brookfield.

FIGURE 3.1


PARTICLE SIZE DISTRIBUTIONS
PERCENT OF TOTAL SOLIDS IN SIZE GROUP
25
20
15
10
5
0
< 63
63 125
125 250
250 500
500 1000
1000 2000
2000 6370
> 6370
SMOOTH
INTER.
ROUGH TEXTURE
SIZE RANGE IN MICRONS

3.3 STREET DIRT CHEMICAL CHARACTERISTICS

As described in Section 2.4, the street dirt samples were chemically analyzed and loading conditions for each constituent were calculated. This section summarizes the observed concentrations and loadings of these constituents.

Table B.1 lists the average observed constituent concentrations for each test area. The weighted average total street dirt concentrations are also shown and are from separate chemical analyses of reconstituted samples. The data from each site was compared to specific site characteristics to determine possible factors that influenced the observed concentrations.

Figures B.1 through B.8 are bar graphs showing particle size constituent concentrations for the test sites. It was found that none of the concentrations of the constituents varied significantly between different test sites, except for lead. The lead data is shown grouped into three categories, smooth streets with heavy traffic (the downtown sites of Fifth St. and the CBD), areas having light traffic but in new residential developments (Tallman, Lepori, and Brookfield), and the other Reno and Sparks sites (Parkview, Mill, Trainer, Locust, and Glendale) that are representative of most of the Truckee Meadows basin. The downtown sites have much greater lead concentrations in all street dirt size categories than the other sites, most probably due to their greater traffic volumes and low street dirt loadings. The new residential areas have light lead street dirt concentrations probably because of their low traffic volumes and high street dirt loadings.

The differences in concentrations as a function of particle size are clearly shown in Figures B.1 through B.8. Chromium, lead, and zinc have their greatest concentrations associated with the smallest particle sizes, as is common for most heavy metals at other study locations. Almost all of the cadmium concentrations were less than the detection limit of three mg/kg, except for some samples collected in the CBD area. Nutrients and COD also showed their highest concentrations in the smallest particle sizes, but the larger particle sizes also had high concentrations, especially COD. Large pieces of organic material (leaves, etc.) would substantially increase these concentrations.

Table A.1 shows the calculated loadings for each constituent for each collected sample, based on reconstituted whole sample analyses and the measured total solids loadings. Table A.2 summarizes these loading values for each test area, sampling period, and if the sample was collected from the whole width of the street or only the driving lane. The first period of sample collection had greater loadings for almost all constituents and samples. The differences were generally about 25 percent. The driving lane loadings were usually less than the near curb loadings. For test areas having lipped street/gutter interfaces or smooth street textures, most of the material was near the curb. Rough streets, however, had most of the material in the driving lanes. Lead was mostly in the driving lanes, even for smooth streets (except those with lipped interfaces), due to the strong association of lead with vehicle activity. Total solids loading relationships were similar to lead, but were not as pronounced. The nutrients and

COD loading relationships varied, but they were mostly found near the curbs for smooth streets, indicating off-street sources.

Table B.2 summarizes the observed whole sample constituent concentrations for each test area, sampling period, and for driving lane samples. The concentrations were similar for both sampling periods. The concentration differences between driving lane and whole street samples were much greater than for the loading differences. The lead concentrations were much greater for the driving lane samples than the whole street samples, except for low traffic, rough textured streets.

Figures B.9 through B.18 show how the concentrations of lead and COD varied with time during the sampling period. Reconstituted whole street width and driving lane sample results are shown for both sampling periods, when available. Figure 3.2 is a figure for lead, showing the lead concentration variations during the first sampling period for five sites. Brookfield and Tallman had low lead concentrations that did not vary much. Trainer, Mill, and Locust had lower total solids loadings and higher lead concentrations. Lead concentrations for these sites appeared to slightly decrease with time during the first period.

Table 3.7 compares the Reno and Sparks street dirt constituent concentrations with values obtained elsewhere. Cadmium, lead, zinc, COD, and phosphorus concentrations are within the range reported elsewhere. Chromium and Kjeldahl nitrogen concentrations are more than ten times less than values obtained elsewhere. The reasons for these discrepancies are not known.

FIGURE 3.2

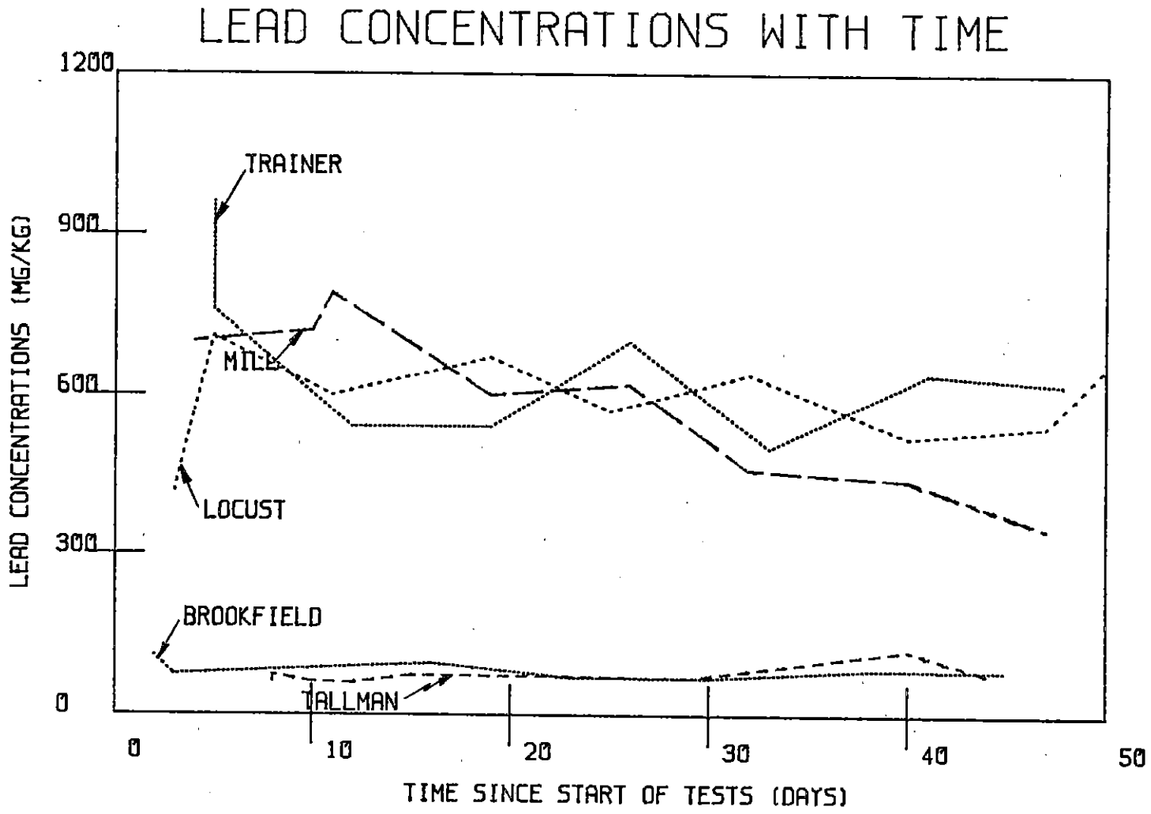

LEAD CONCENTRATIONS WITH TIME
1200
900
600
300
0
LEAD CONCENTRATIONS (MG/KG)
TRAINER
MILL
LOCUST
BROOKFIELD
TALLMAN
0
10
20
30
40
50
TIME SINCE START OF TESTS (DAYS)

Table 3.7

RENO/SPARKS STREET DIRT CONSTITUENT CONCENTRATIONS
COMPARED WITH DATA FROM OTHER LOCATIONS (mg/kg)

Constituent	Range of Reno/Sparks Values	Typical Reno/Sparks Values	San Jose[1]			Castro Valley	Bellevue, Washington
			Smooth Asphalt	Poor Asphalt	Oil and Screens	Smooth Asphalt[2]	Smooth Asphalt[3]
Cadmium	<3	<3	2	3	1	-	1
Chromium	15→60	30	450	450	350	200	200
Lead	75→3,500	100→2,500	5,500	2,000	1,000	1,600	1,000
Zinc	100→400	200	750	500	250	200	200
COD	70,000→150,000	100,000	120,000	110,000	80,000	90,000	130,000
Phosphorus	700→1,000	800	-	-	-	500	500
Nitrate-N	10→30	25	-	-	-	-	-
Nitrite-N	2→30	5	-	-	-	-	-
Kjeldahl-N	60→200	150	2,000	2,300	1,000	1,600	1,500

[1] Pitt, 1979.
[2] Pitt and Shawley, 1982.
[3] Pitt and Sutherland, 1981.

433.017.1

Table 3.7 (continued)

Constituent	Range of Reno/Sparks Values	Typical Reno/Sparks Values	Ottawa, Ontario[4] Smooth & Moderate Asphalt	Ottawa, Ontario[4] Rough Asphalt	Ottawa, Ontario[4] Very Rough Asphalt	Phoenix[5]	Milwaukee[5]	Bucyrus, Ohio[5]	Baltimore[5]	Atlanta[5]	Tulsa[5]
Cadmium	<3	<3	-	-	-	4	3	3	8	1	2
Chromium	15→60	30	-	-	-	200	150	200	300	200	100
Lead	75→3,500	100→2,500	1,000	400	250	1,000	800	900	2,000	700	700
Zinc	100→400	200	-	-	-	300	300	250	600	300	200
COD	70,000→150,000	100,000	-	-	-	50,000	20,000	20,000	20,000	30,000	100,000
Phosphorus	700→1,000	800	-	-	-	-	-	-	-	-	-
Nitrate-N	10→30	25	-	-	-	-	-	-	-	-	-
Nitrite-N	2→30	5	-	-	-	-	-	-	-	-	-
Kjeldahl-N	60→200	150	-	-	-	2,800	500	900	2,000	1,000	2,000

[4] Pitt, 1982.
[5] Sartor and Boyd, 1972, and Pitt and Amy, 1973.

433.017.2

3.4 STREET DIRT ACCUMULATION CHARACTERISTICS

The loading and accumulation observations shown in Table A.1 were analyzed to estimate accumulation rates for the test sites. The accumulation periods shown in this table are the number of days since a significant rain or since the area was cleaned. A significant rain is usually greater than about 1/5 inch over several hours. This rain can remove almost all of the street dirt that is likely to be removed by rains. The accumulation periods ranged from less than 0.1 days for samples collected immediately after street cleaning to about 45 days.

Figures D.1 through D.10 show the actual data along with the estimated accumulation curves. Separate accumulation curves were prepared for each of the sites for the two test periods, when possible. The CBD and Lepori sites had the same accumulation curves for both test periods. Parkview, Fifth, and Glendale had substantially different accumulation curves for the test periods. The first period was much windier (as described in Section 3.1) and had higher street dirt loadings.

Figure 3.3 shows all the accumulation curves together, grouped by influencing factors. These factors are street texture, street/gutter interface condition, and wind condition. The curves are almost parallel, but with significant differences in their initial loading values. The accumulation rates from about one through six weeks are close, but the initial accumulation rates (the deposition rates) vary more. The rougher streets have greater initial loading values (more material on the streets that cannot be removed by rains). New residential areas have high loadings probably because of close-by soil sources eroding during construction and landscaping activities. The first period, windy, values are greater also because of probable greater erosion contributions.

Table D.1 presents the calculated loading values and deposition values for all of the constituents, along with total solids. The loadings obviously increase with time in the absence of natural or other removal mechanisms. The rate of accumulation, however, decreases with time. The initial accumulation rates are as much as ten times the accumulation rates observed near the end of the time period studied. The deposition rates are expected to be fairly constant with time, except for natural conditions (such as winds or vegetation) that change more slowly. The differences in the deposition rates and the accumulation rates at any one time is most likely due to fugitive dust losses from the street surface. These fugitive losses increase as the street loadings increase (with constant street textures) and with increased winds (either natural or traffic induced). The actual fugitive dust losses from the street surfaces can be calculated by subtracting these two rates. Frequent street cleaning may therefore play an important role in reducing fugitive dust losses. The largest fugitive dust losses occur after about one to two weeks of street dirt accumulation. Therefore, effective street cleaning at least every two weeks should be sufficient to control these losses. Small street cleaning frequency changes, especially if the cleaning frequency is already less than one week would probably not be significant.

FIGURE 3.3

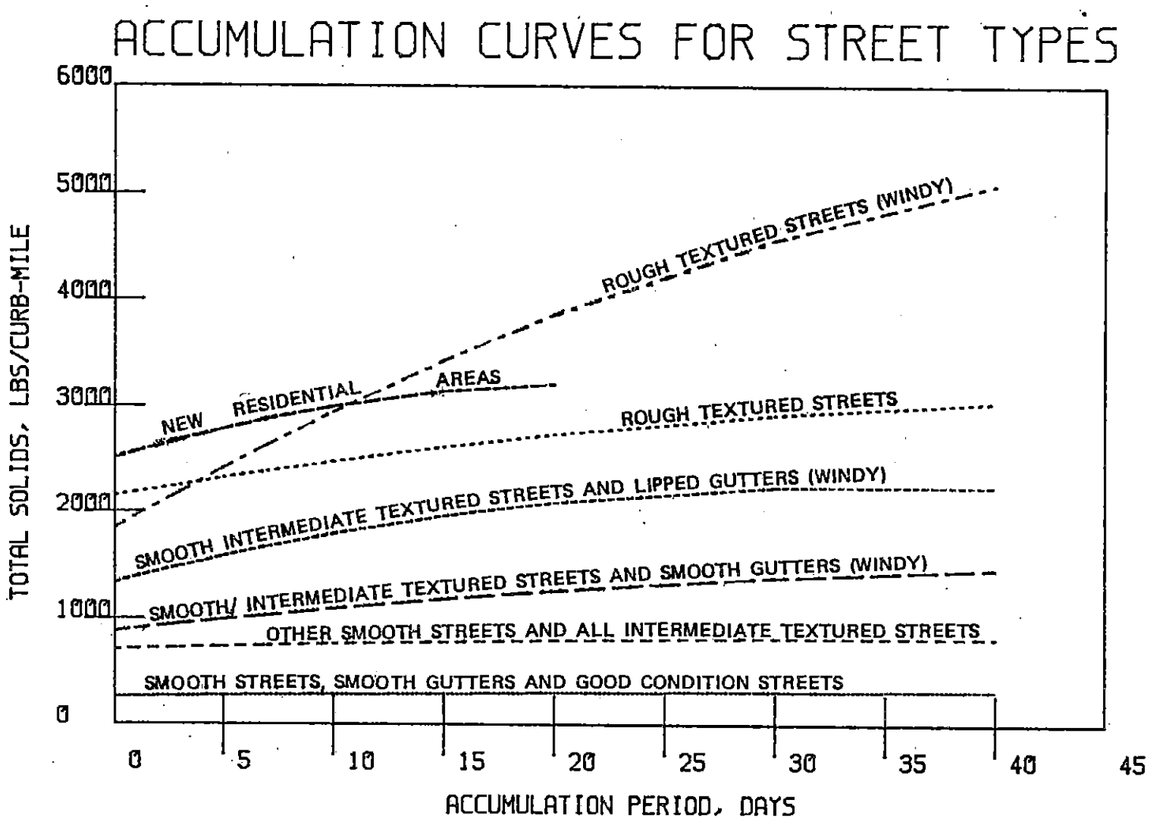

ACCUMULATION CURVES FOR STREET TYPES
TOTAL SOLIDS, LBS/CURB-MILE
6000
5000
4000
3000
2000
1000
0
ROUGH TEXTURED STREETS (WINDY)
NEW RESIDENTIAL AREAS
ROUGH TEXTURED STREETS
SMOOTH INTERMEDIATE TEXTURED STREETS AND LIPPED GUTTERS (WINDY)
SMOOTH/ INTERMEDIATE TEXTURED STREETS AND SMOOTH GUTTERS (WINDY)
OTHER SMOOTH STREETS AND ALL INTERMEDIATE TEXTURED STREETS
SMOOTH STREETS, SMOOTH GUTTERS AND GOOD CONDITION STREETS
0
5
10
15
20
25
30
35
40
45
ACCUMULATION PERIOD, DAYS

Table 3.8 compares the initial total solids loading and deposition values estimated for the seven Reno/Sparks conditions with values obtained elsewhere. When street texture is considered, the Reno and Sparks values are comparable to the other data. The rough streets under windy conditions had deposition values much greater than found elsewhere. The smooth street values were less than found elsewhere. The accumulation values are all much closer to values from other locations. In all cases, rough streets have much greater initial loading values than smooth streets, indicating the ability of rain and street cleaning equipment to keep smooth streets much cleaner than rough streets.

Table 3.8

RENO/SPARKS STREET DIRT DEPOSITION RATES
COMPARED WITH DATA FROM OTHER AREAS

	Initial Loading Value (lb/curb-mi)	Deposition (Initial Accumulation) Rate (lb/curb-mi/day)
Reno/Sparks		
Smooth streets and gutters in good condition	270	2.6
Other smooth streets and intermediate streets	710	6.1
Rough streets	2,200	36
New residential areas	2,500	61
Smooth and intermediate streets with smooth gutters (windy)	880	24
Smooth and intermediate streets with lipped gutters (windy)	1,300	53
Rough streets (windy)	1,900	120
San Jose[1]		
Smooth asphalt, good condition	130	15
Smooth asphalt, fair to poor condition	290	15
Rough asphalt, poor condition	780	20
Oil and screens	1,800	20
Castro Valley[2]		
Smooth asphalt	300	40
Bellevue[3]		
Smooth asphalt	380	13
Nationwide[4]		
Residential (smooth asphalt/good to fair)	400	20
Industrial (rough asphalt/poor)	670	40
Commercial (smooth asphalt/good)	300	15

[1] Pitt, 1979.
[2] Pitt and Shawley, 1982.
[3] Pitt and Sutherland, 1981.
[4] Sartor and Boyd, 1972, and Pitt and Amy, 1973.

3.5 STREET CLEANER PRODUCTIVITY

The street cleaning test program was described in Section 2.2 . The tests included measuring street dirt loadings immediately before and after the streets were cleaned in a variety of ways. The most important tests included investigating street cleaner performance in the manner that the equipment is currently being used: a single pass next to the curb. Other special tests were conducted to investigate possible changes to the existing street cleaning program. These tests included making several passes in a short period of time, changing the street cleaner vehicle speed, using different types and combinations of equipment, and cleaning the full width of the street. The results of these tests are described in this section.

Single Street Cleaner Passes Next to the Curb

Most of the street cleaner testing effort involved single passes that were made next to the curb. Thirteen tests were conducted in five test areas during the start of the second sampling period. Two of these sites were in areas having rough streets (Lepori and Glendale), and three of the sites had smooth textured streets (the CBD, Parkview, and Fifth).

Figure 3.4 shows the resultant productivity curves for these two groups, expressed as the expected residual loading for the given initial loading. Table 3.9 shows the straight line curve coefficients for the eight different particle sizes and for the total material. The regression coefficients were better for the smooth street group, which were usually greater than 0.85, than for the rough street group, which were about 0.75 or less. The smooth streets were capable of being cleaned to lower residual loading values than the rough streets, about 200 verses 2000 lbs/curb-mile. If the initial loading values were close to these critical values, the amount of material removed from the streets during the cleaning operations would be very little. Very frequent street cleaning does not result in significatly cleaner streets than more moderate cleaning frequencies. As an example, cleaning a street three times a week results in streets that are just about as clean as if they were cleaned daily.

The percentage removal values vary substantially, depending on the initial loading conditions and the street texture. If the initial loading was 4500 lbs/curb-mile on a rough street, the resultant residual loading would be about 2800 lbs/curb-mile, with a removal of about 1700 lbs/curb-mile, or 38 percent. If the initial loading value was only 2000 lbs/curb-mile, the residual loading value would be about the same, with no removal. If the initial loading was less than 2000 lbs/curb-mile for rough textured streets, the percentage removal could be negative, due to erosion of the street surface, or blowing material onto the street from adjacent areas. This has been observed in many other studies.

If an initial loading value of 1200 lbs/curb-mile occurred on a smooth street, the residual loading value would be about 900 lbs/curb-mile. The removal would be about 300 lbs/curb-mile, or 25 percent. If the initial load was 500 lbs/curb-mile, the residual load would be about 300 lbs/curb-mile, with a removal of 200 lbs/curb-mile, or about 40 percent.

FIGURE 3.4

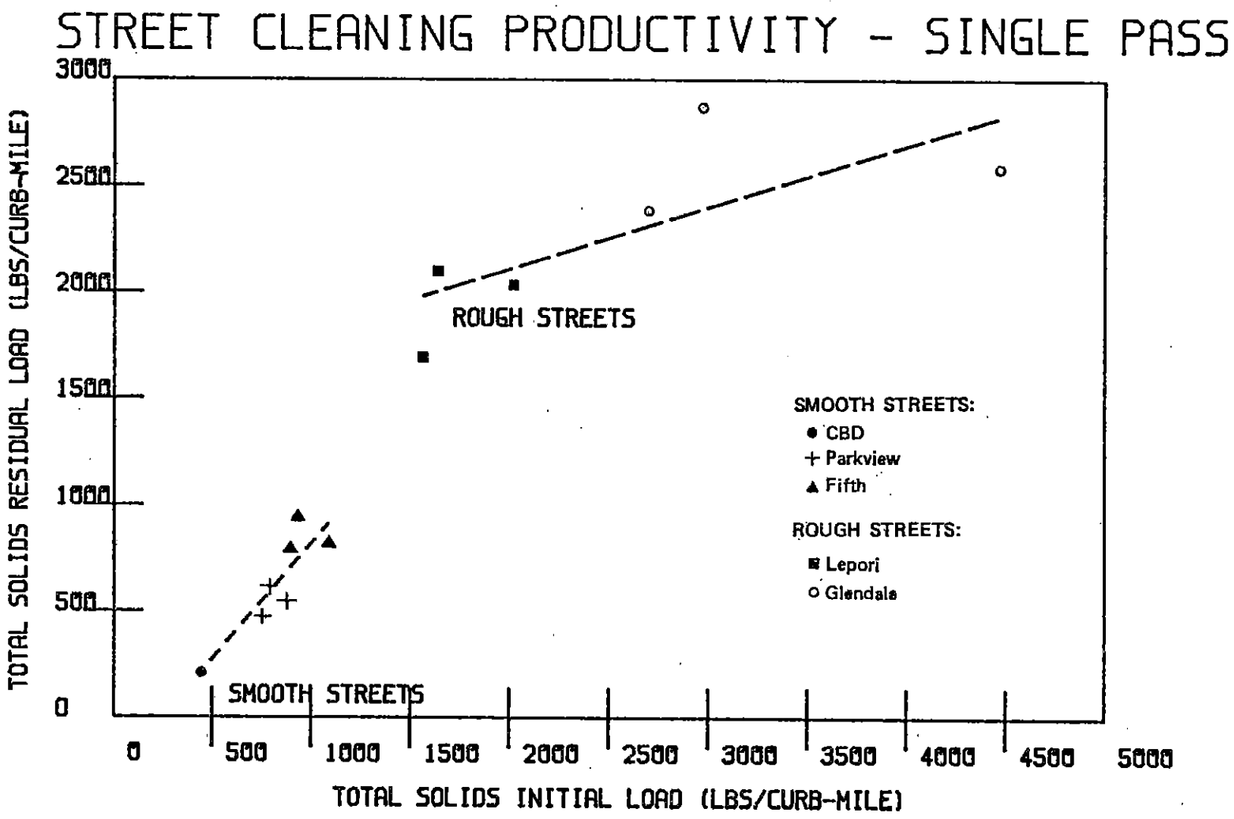

STREET CLEANING PRODUCTIVITY - SINGLE PASS
TOTAL SOLIDS RESIDUAL LOAD (LBS/CURB-MILE)
3000
2500
2000
1500
1000
500
0
ROUGH STREETS
SMOOTH STREETS
SMOOTH STREETS:
CBD
Parkview
Fifth
ROUGH STREETS:
Lepori
Glendale
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
TOTAL SOLIDS INITIAL LOAD (LBS/CURB-MILE)

Table 3.9

STREET CLEANER PRODUCTIVITY-SINGLE PASS
EQUATION COEFFICIENTS[1] AND DATA RANGE

Particle Size	Smooth Streets			Rough Streets		
Range (microns)	Inter-cept	Slope	Range for x	Inter-cept	Slope	Range for x
>6,370	5.8	0.66	0.1→40	22	0.49	10→200
2,000→6,370	-16	1.12	20→70	210	0.21	160→610
1,000→2,000	-1.0	0.75	30→100	270	0.38	310→700
500→1,000	-34	1.02	80→180	270	0.44	380→830
250→500	-54	1.06	110→250	260	0.27	280→730
125→250	-81	1.17	100→270	200	0.22	180→600
63→125	-47	1.13	60→160	120	0.25	95→400
<63	17	0.53	20→100	130	0.13	75→400
Total	-270	1.08	440→1,090	1,530	0.29	1,570→4,470

[1] Y = A + BX
Y = Residual Total Solids Load (lb/curb mile)
A = Intercept
B = Slope
X = Initial Total Solids Load (lb/curb mile)

Multiple Pass Street Cleaning Productivity

Three passes within three days (many sites had all three passes on the same day) were made at all test sites at the beginning of both test periods. Figure 3.5 shows the results of this multiple pass effort. Little difference between sites having different textures could be found due to the data scatter. Therefore, all site data is combined for this analysis. This curve overlaps the single pass removal curves shown in Figure 3.4, with little significant difference. Therefore little, if any, additional improvement can be expected for multiple passes in a short period of time when compared to a single pass. Extensive cleaning efforts (ten passes each week) were monitored in San Jose (Pitt, 1979), with resultant minimum street dirt loadings maintained over the period of cleaning. Very small differences were found for rough textured streets, but extensive cleaning had more noticeable effects for smooth streets. Three passes in a short period of time (less than three days) were also found to result in close to minimum loadings.

Speed Effects on Street Dirt Removal

Several street cleaner tests were conducted at the Glendale, Brookfield and Lepori sites at vehicle operating speeds of about 2, 4, 8, and 15 miles per hour. Figures 3.6, C.1 and C.2 show the results of these tests. Four miles per hour results in the lowest residual loads for these streets at initial loadings above 1500 lbs/curb-mile. At lower initial loadings for these intermediate to rough textured streets, the vehicle speed was not as important. The fastest vehicle speeds of 13 to 15 mph were found to remove appreciable material from the streets, but the removed material probably did not end up in the vehicle hopper: it was resuspended as fugitive dust by the turbulence of the fast moving vehicle. The dust clouds behind fast moving street cleaning equipment is quite noticeable.

The removal of street dirt was much more pronounced at the lower speeds at the Brookfield site, as shown in Figures C.1 and C.2, because of the greater initial loadings. The removal was very little at Brookfield for the higher vehicle speeds. This also shows the importance of lower vehicle speeds for very dirty roads.

The Effects of Different Street Cleaner Types

All of the street cleaner tests were conducted using a Mobil four wheel mechanical street cleaner. Some tests were also conducted at the Glendale site using an Ecolotec four wheel vacuum assisted mechanical street cleaner. Table 3.10 shows the results of these tests. The Ecolotec cleaned two test sections, while the Mobil cleaned three sections, including one of the Ecolotec sections a second time. The initial loading values were all very high for all of these tests, and the reported removal values were similar. These limited test results do not show any significant differences between the two equipment types for these street and loading conditions. The multiple pass test result was similar to the single pass test results for each equipment type individually. Extensive tests conducted in San Jose (Pitt, 1979) and Castro Valley (Pitt and Shawley, 1982) to compare different equipment types (Mobil, Ecolotec, and Tymco) also found little practical differences in the performance of different street cleaners.

FIGURE 3.5

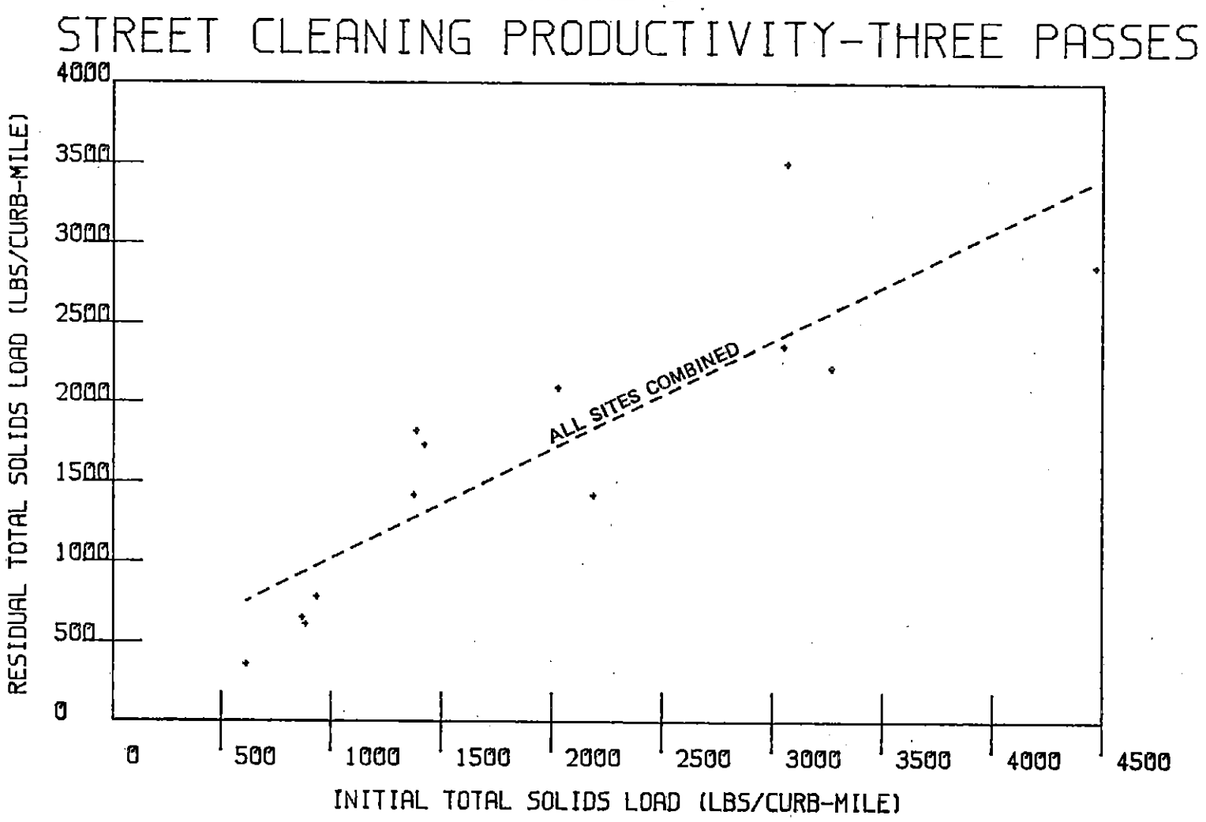

STREET CLEANING PRODUCTIVITY-THREE PASSES
RESIDUAL TOTAL SOLIDS LOAD (LBS/CURB-MILE)
4000
3500
3000
2500
2000
1500
1000
500
0
ALL SITES COMBINED
0
500
1000
1500
2000
2500
3000
3500
4000
4500
INITIAL TOTAL SOLIDS LOAD (LBS/CURB-MILE)

FIGURE 3.6

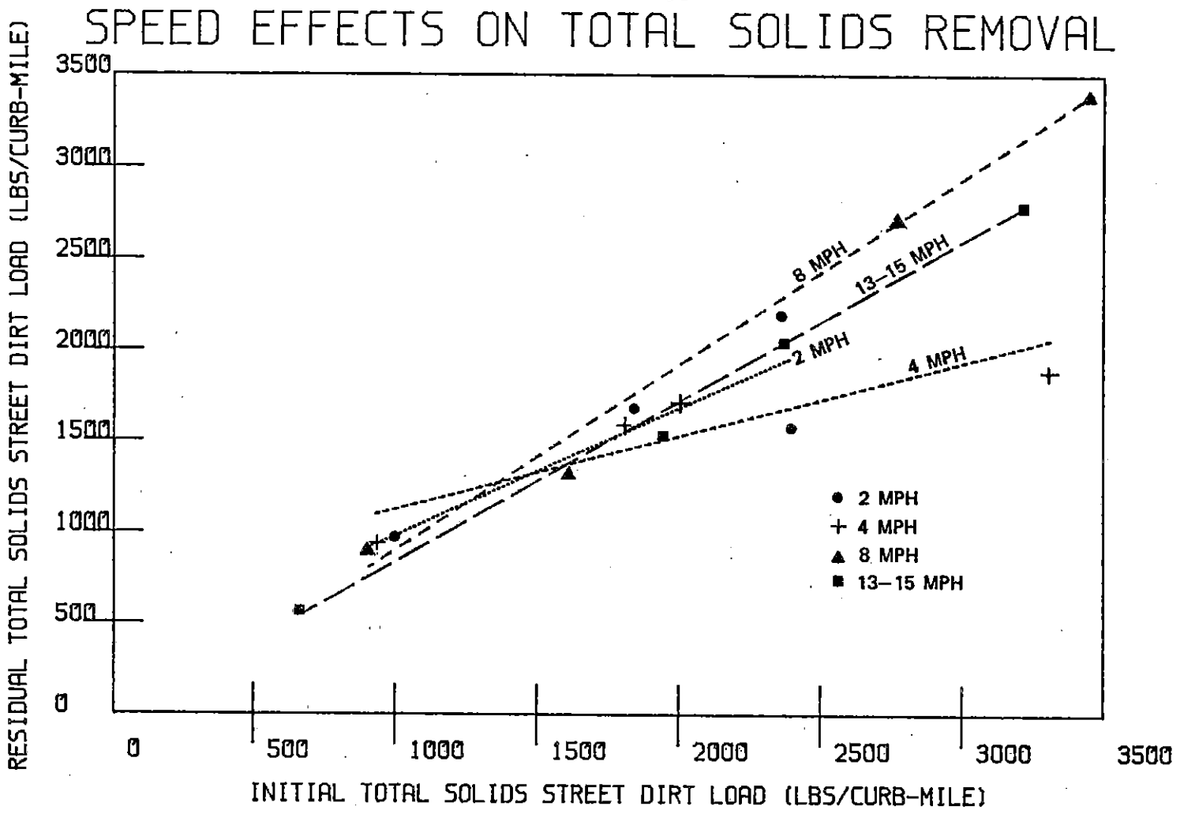

SPEED EFFECTS ON TOTAL SOLIDS REMOVAL
RESIDUAL TOTAL SOLIDS STREET DIRT LOAD (LBS/CURB-MILE)
3500
3000
2500
2000
1500
1000
500
0
0
500
1000
1500
2000
2500
3000
3500
INITIAL TOTAL SOLIDS STREET DIRT LOAD (LBS/CURB-MILE)
8 MPH
13-15 MPH
2 MPH
4 MPH
2 MPH
4 MPH
8 MPH
13-15 MPH

Table 3.10

EFFECTS OF DIFFERENT EQUIPMENT TYPES
USED AT THE GLENDALE SITE
(lb/curb mile)

Equipment	Initial Load	Residual Load	Amount Removed	Percent Removed
Ecolotec	7,107	4,729	2,378	33
Mobil and Ecolotec	7,107	4,458	2,649	37
Mobil	3,336	1,997	1,339	40
Mobil	6,806	3,881	2,925	43

Driving Lane Load Changes due to Street Cleaning

Six driving lane samples were collected before and after curb street cleaning in some of the second period sampling locations. Past studies have indicated the redistribution of street dirt across the street during cleaning. The heavy loads near the curb are brushed towards the main pick-up broom, with some of the material landing in the driving lanes. Figure 3.7 shows that the driving lane loadings were increased slightly (about five percent) after street cleaning.

Across the Street Redistribution of Street Dirt During Street Cleaning

As noted above, street dirt is brushed away from the curb during street cleaning, with some of the material landing outside of the cleaning path. Other redistributed street dirt is not picked up by the main pick-up broom. Figures 3.8, C.3, C.4 and C.5 show how the street dirt was redistributed during cleaning during special tests conducted at the Glendale, Brookfield, and Lepori test sites. These streets were rough, with appreciable initial loadings in the driving lanes (about 40 percent of the total street loadings were in the 16 to 25 foot section from the curb). The center of the streets were also cleaned for some of these tests, with resultant blowing of street dirt back towards the curb.

Figure 3.9 compares full width street cleaning with curb side cleaning for these rough streets. There was little difference noted between these cleaning efforts. The center of the street loadings for these rough streets were fairly high, but was still below the loading values needed for positive street dirt removal for rough streets. The center of the street loadings for smooth streets were much lower and would also be too low for positive removals. The street dirt in the center of the streets does not appear to be available for removal, either by street cleaning, winds, or rains. Powerful street flushing could remove this material, but at great use of water and with apparently little urban runoff or air pollution benefit.

FIGURE 3.7

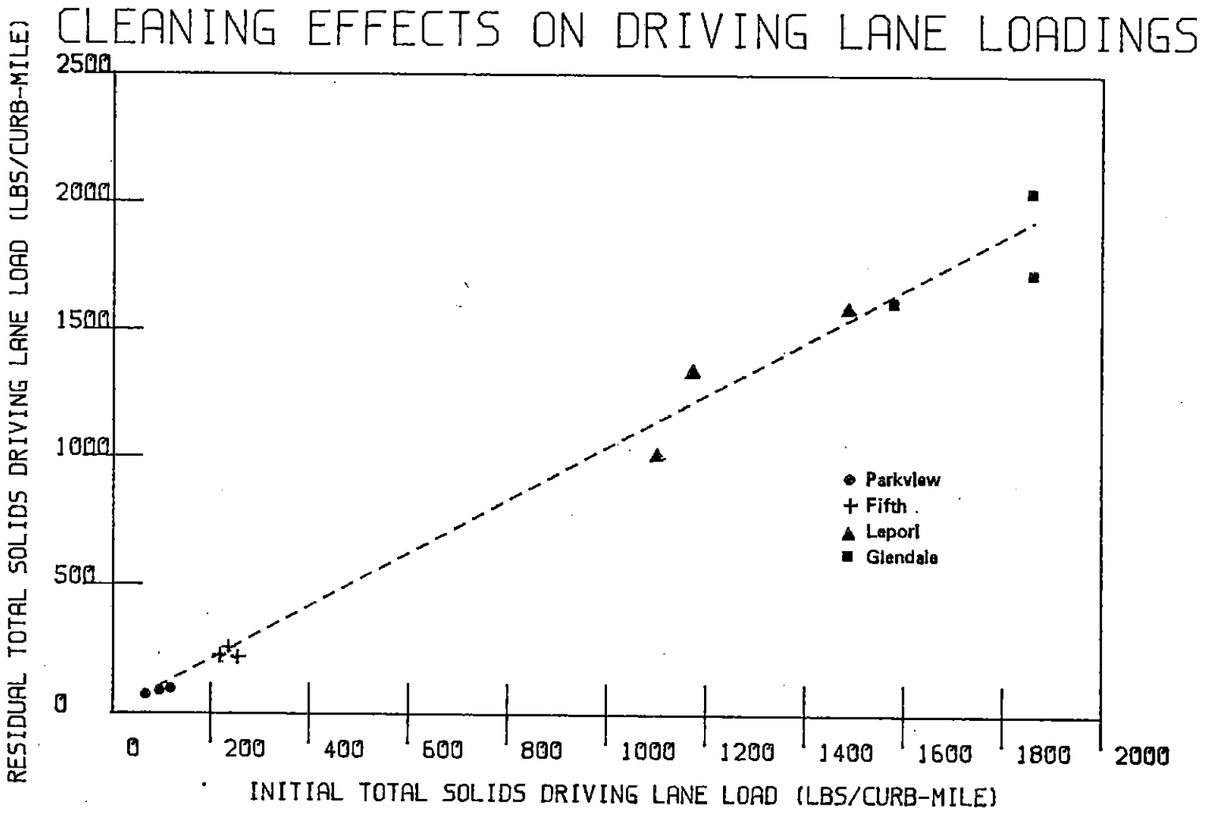

CLEANING EFFECTS ON DRIVING LANE LOADINGS
RESIDUAL TOTAL SOLIDS DRIVING LANE LOAD (LBS/CURB-MILE)
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
1400
1600
1800
2000
INITIAL TOTAL SOLIDS DRIVING LANE LOAD (LBS/CURB-MILE)
Parkview
Fifth
Leporl
Glendale

FIGURE 3.8

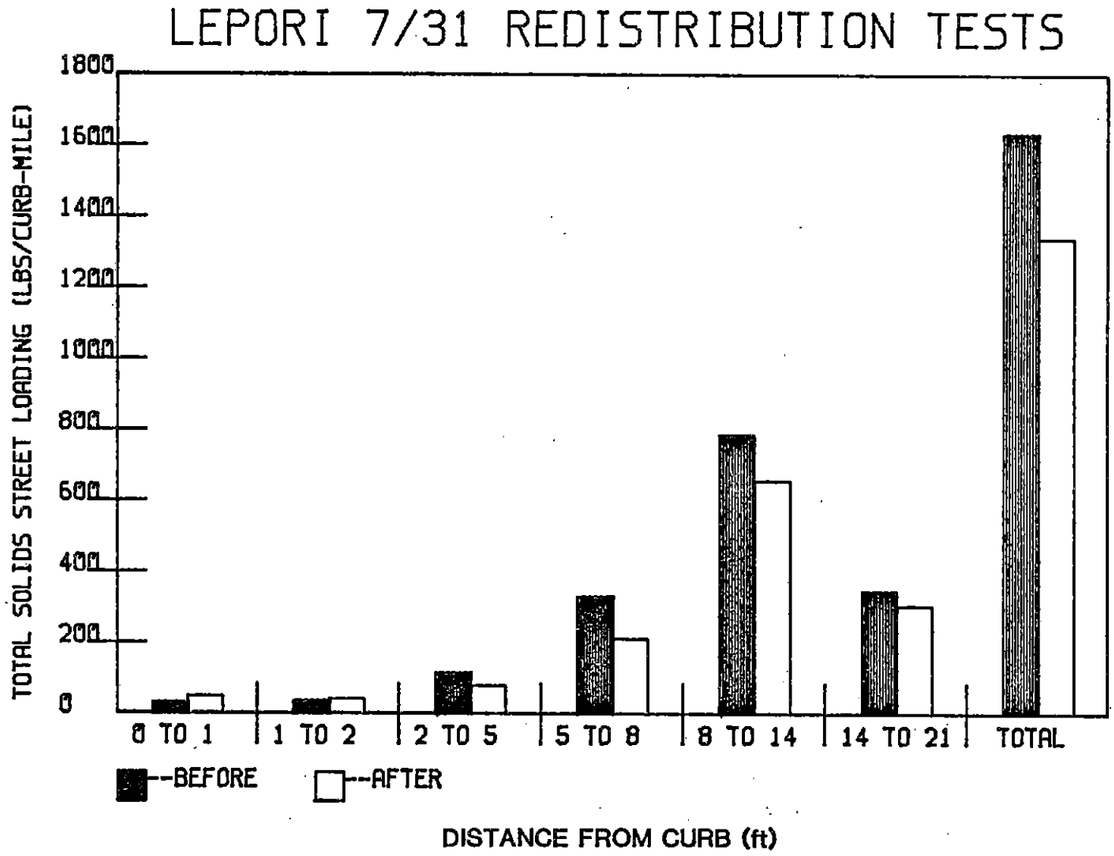
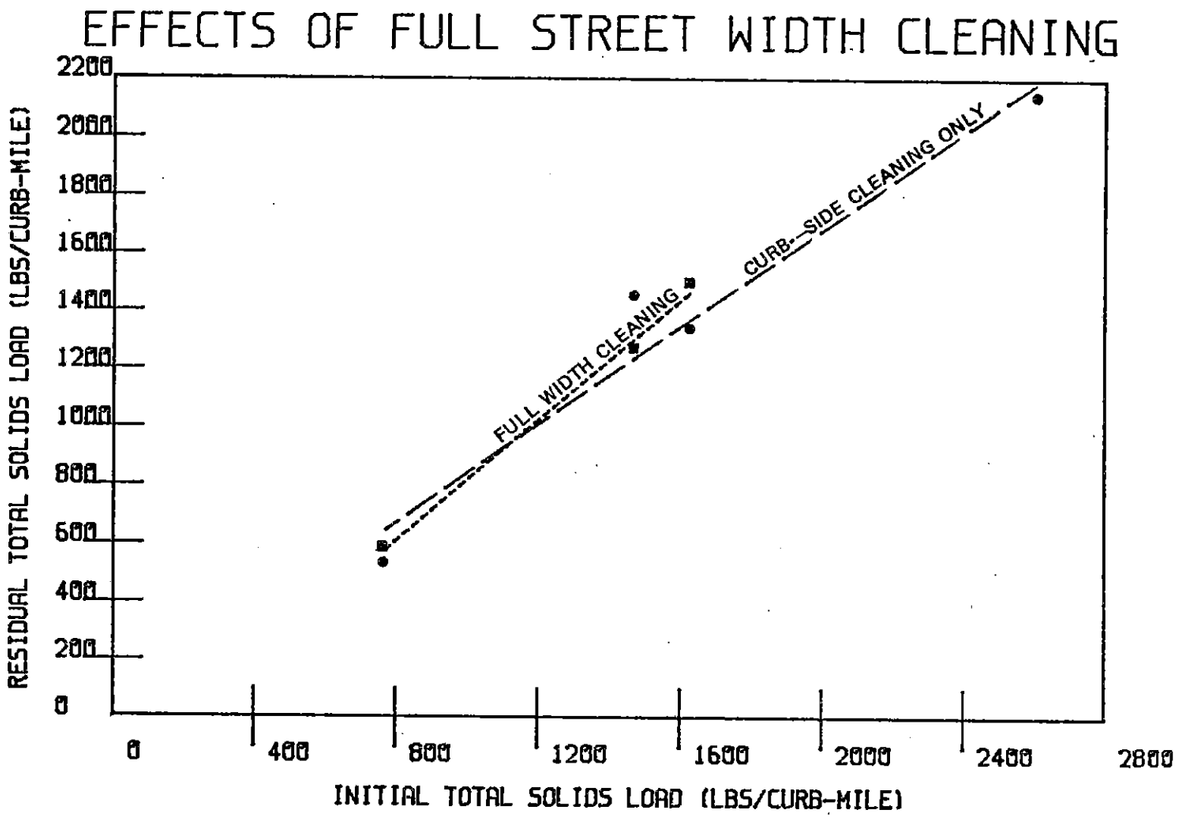

LEPORI 7/31 REDISTRIBUTION TESTS
TOTAL SOLIDS STREET LOADING (LBS/CURB-MILE)
1800
1600
1400
1200
1000
800
600
400
200
0
0 TO 1
1 TO 2
2 TO 5
5 TO 8
8 TO 14
14 TO 21
TOTAL
--BEFORE
--AFTER
DISTANCE FROM CURB (ft)

FIGURE 3.9


EFFECTS OF FULL STREET WIDTH CLEANING
RESIDUAL TOTAL SOLIDS LOAD (LBS/CURB-MILE)
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
FULL WIDTH CLEANING
CURB-SIDE CLEANING ONLY
0
400
800
1200
1600
2000
2400
2800
INITIAL TOTAL SOLIDS LOAD (LBS/CURB-MILE)

Chapter 4

Chapter 4
SUMMARY

This report describes the special field studies undertaken as part of the Washoe County Urban Stormwater Management Program and presents the results of these studies. The data collected and reported here can be grouped into two major categories:

- Data on site-specific physical and chemical characteristics of the street particulate material accumulating in the Truckee Meadows area.
- Data on the site-specific street-cleaner performance characteristics throughout the Truckee Meadows area.

A tremendous amount of data and information is presented in this report.

The data analyses chapter (Chapter 3) was not intended to be the final word on the data's significance. Subsequent analyses will be required to calibrate the stormwater quality model, to use the model to project areawide loadings under various conditions, and to develop recommendations on public works practices such as street cleaning.

Based on the analyses conducted and reported to date, several statements can be made about the characteristics of the street dirt and the factors that influence it in the Truckee Meadows study area.

Climatic Conditions

- 1981 (the year of the field data collection) was an exceptionally dry year. No rain was recorded during June and July.
- High winds experienced in July 1981 during the field studies were unusual. They appear to have affected the observed loading values. This may have been caused by fugitive dust source contributions to the street loads.

Particle Size Distributions

- Street texture appears to have the greatest influence on particle size distribution. Rougher streets have a higher concentration of larger particles than intermediate or smoother streets.
- Driving lane samples always had larger median particle sizes than corresponding full street width samples; therefore, street dirt near the curb is finer than that found in the driving lane.
- The observed median particle sizes for the Truckee Meadows data were similar to those reported elsewhere for rough streets.

Chemical Characteristics

- None of the concentrations of the various chemical constituents varied significantly between different test sites, except for lead.
- The commercial sites had much higher lead concentrations mostly because of their higher traffic volumes and low street dirt loadings.

- The new residential areas had much lower lead concentrations probably because of their low traffic volumes and high street dirt loadings.

- The differences in chemical concentrations as a function of particle size were substantial as expected. However, their variations were similar to data collected elsewhere.

- Lead was mostly found in the driving lane even though the total solid loadings were generally greater in the curb lane. This demonstrates the strong association of lead with vehicle activity.

- Cadmium, lead, zinc, COD, and phosphorus concentrations were within the range reported elsewhere. Chromium and Kjeldahl nitrogen concentrations were more than ten times less than values obtained elsewhere. The reason for these discrepancies is not known.

Accumulation Characteristics

- Street texture, street/gutter interface, and wind conditions appear to have the greatest influence on street particulate accumulation characteristics.

- The rougher streets have the greatest initial loading values. New residential areas have high loadings because nearby soil sources erode during construction and landscaping activities.

- The first phase of data collection had much greater loading values primarily from high winds that increased fugitive dust contributions.

- The actual loading values increase with time in the absence of natural or other removal mechanisms.

- The rate of accumulation decreases with time. The initial accumulation rates (i.e., deposition rates) are as much as ten times greater than the accumulation rates observed near the end of the time period.

- The deposition rates are expected to be fairly constant with time; however, the rate of deposition will be influenced by natural conditions such as wind.

- The differences in the deposition rates and accumulation rates at any one time are probably caused by fugitive dust losses from the street surface, which will increase as loadings increase.

- The largest fugitive dust losses occur after about one to two weeks of street dirt accumulation. Therefore, effective street cleaning at least every two weeks should be sufficient to control these losses.

- Driving lane loadings were usually less than curb lane loadings. For test areas having lipped street/gutter interfaces or smooth street textures, most of the material was near the curb. On rough streets, however, most of the material was in the driving lane.

- The Truckee Meadows accumulation characteristics are comparable to those observed elsewhere, provided similar street textures are considered. The rough street under windy conditions had deposition rates (i.e., initial accumulation rate) much greater than found elsewhere. The smooth street deposition rates were less than those found elsewhere. The accumulation rates are all much closer to those values found at other locations in the country. Thus, rough streets are acting as traps preventing the loss of material to wind, whereas this was not the case for smooth streets.

Street Cleaner Productivity

- The street cleaner productivity relationships found in the Truckee Meadows area were similar to those found elsewhere.

- Percentage removal values vary substantially depending on the initial loading conditions and street texture. If the initial loading was less than 2,000 pounds per curb mile for rough textured streets, the percentage removal could be negative because of street surface erosion. The greatest removals were found on highly loaded smooth to intermediate textured streets.

- Little additional improvement in removal percentage can be expected for multiple passes rather than single passes in a short period of time.

- A street cleaner speed of 4 miles per hour resulted in the lowest residual loadings for initial accumulation values of 1,500 pounds per curb mile or greater. At lower initial loadings, vehicle speed was not important because of the rough street texture.

- The influence of vehicle speed in the removal of street dirt was much more pronounced at the Brookfield site since the initial loadings were very high.

- Limited tests conducted on the Glendale site found no significant differences in the performance of the two different equipment types that were tested. Extensive tests conducted in San Jose and Castro Valley, California, found few practical differences in the performance of different types of street cleaners.

- Some material was redistributed across the street as the result of curb lane sweeping. This is consistent with results obtained in other studies.

- Little difference was noted between full-width street cleaning and curb lane cleaning only. Although the driving lane loadings were relatively high, they still remained lower than those needed for positive removals considering the different street texture.

- The street dirt located in the driving lane does not appear to be available for removal, either by street cleaning, winds, or rains. Powerful street flushing could remove this material, but only by using a large amount of water and with apparently little urban runoff or air pollution benefit.

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